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Sighteo: Navigate by Sight

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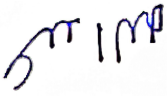
Signature: 

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Abstract

Vision impairment causes significant navigation challenges for millions of people around the world, leading to reduced autonomy and increased safety risks. Sighteo's plan is to meet this demand through a mobile application, which will be based on the latest object detection technologies to instantly spot barriers and clear paths. Major project tasks include gathering an appropriate dataset or adapting an existing one, choosing a model that would unite speed and accuracy in the right proportions, and developing a user-friendly visual interface that would communicate directions through audio and tactile feedback. Testing will be performed in controlled environments to refine the reliability and usability of the model. Sighteo's aim at the project conclusion is to deliver a navigation tool that is not only functional but also easy to use and thus, giving visually challenged users more freedom and assurance.

Executive Summary

The main focus of this project is to research the datasets, models and limitations of existing assistive technologies for visually impaired individuals to develop an AI-based navigation system that overcomes the limitations of existing solutions. Our fine-tuned model will be enclosed in a thoughtfully designed React Native mobile application, specific to the needs of the visually impaired individuals, which will make use of the smartphone camera to detect obstacles in real-time and guide users on safe paths via audio or haptic feedback. Through this project, We aim to overcome limitations of existing technologies that require, for example, expensive hardware like LiDAR, ultrasonic sensors, mixed-reality headsets or human volunteers thus sacrificing autonomy, affordability, and scalability.

A thorough literature review established the foundation for this work. Previous studies and applications demonstrated promising advances in object detection and pathfinding for assistive navigation but were found lacking in real-world usability, affordability, and continuous guidance. Research papers like MR.NAVI and VocalEyes integrate state-of-the-art AI but are founded upon either expensive hardware or mixed reality headsets. Competing applications like Glidance, Be My Eyes, PingPath, and Seeing AI focus upon current industry trends but exhibit continued weaknesses in their incorporation of Artificial Intelligence, with Be My Eyes solely founded upon the implementation of a human guide for the purpose of navigation assistance. This expands the need for a distinct solution which prioritises velocity, precision, and autonomy.

The methodology uses an iterative design process. Mobile integration, training and optimizing YOLO-based object detection models, dataset preparation and augmentation, and a feedback module that transforms detections into useful haptic or audio cues are all included. The scope of the project begins with indoor navigation in controlled environments, with planned extensions toward semi-controlled indoor spaces, dynamic real-world scenarios, and ultimately outdoor navigation.

This study finds its biggest payoff in the society area not just in the academic circles. It is, in fact, likely to be a proper solution for the visually impaired people's everyday difficulties. The project, which bears the stamp of sustainability, perfectly aligns with the goals of the United Nations SDGs (SDG 10: Reduced Inequalities, SDG 11: Sustainable Cities and Communities) as it facilitates the visually impaired to live more independently and be more socially integrated. The product has the potential to generate revenue and therefore will attract investments from the hospitals, NGOs, and accessibility focussed groups.

To sum up, the system that has been suggested demonstrates the way that computer vision, AI, and mobile technologies can be combined to totally get rid of the main drawbacks that have characterized the

existing assistive navigation alternatives. Changing the focus of the project to affordability, accessibility, and real-time usability is a slow but sure way to give the visually impaired the right to safe, independent movements in indoor environments and also prepare the ground for further challenging navigation contexts in the future.

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Chapter 1 Introduction

Millions of people around the world face significant challenges due to visual impairment, making everyday navigation both difficult and unsafe. Walking independently without the aid of eyesight increases the risk of falls, which reduces the confidence in the execution of day-to-day activities. Traditional assistive devices such as walking sticks orhuman guides help only in partial assistance but provide no inherent independence.

Technology has advanced significantly in assistive technologies. We however lack any affordable and pervasive solution which allows the visually impaired individuals to navigate in real time without the assistance of specialized sensors or humans. Solutions we have include the utilization of very high-cost hardware like LiDAR or video call with humans. It does not actually exploit the actual potential of Artificial Intelligence and Object Detection.

In order to bridge this gap, our project, Sighteo, uses developments in computer vision and artificial intelligence to create a mobile application that analyzes the surroundings in real time and provides haptic and audio cues for navigation. The purpose of this project is to investigate the performance of different models and refine them along with specially prepared datasets to meet the requirements of precision and speed together with users' safety, collision-free navigation in indoor places, simplicity, low-cost, and accessibility.

1.1 Purpose of this Document

The purpose of this document is to present the design, development plan, and future of our Final Year Project (FYP), Sighteo. The main objective of the project is to create a mobile-based navigation system that empowers visually impaired individuals to navigate indoor environments independently by providing real-time obstacle detection and path guidance.

The project intends to detect, through research and experimentation, the most accurate and fastest object detection models with the best-suited datasets. The main aim is to ensure the detection of obstacles by the chosen model in real time and to give smooth and uninterrupted feedback to the visually impaired, thus allowing them to use their indoor space with confidence and safety.

In order to reach this goal, the project first includes training dataset curation or adaptation, then experimenting with detection models like YOLO to settle on the most acceptable accuracy or speed compromise, and finally, the integration of the trained model in a mobile application interface. Usability is coming from audio and haptic signals as the feedback method. The controlled environments will be the final testing ground for the system to measure its reliability and responsiveness.

1.2 Intended Audience

The intended audience for this document primarily includes the academic evaluation panel and our project supervisor, who will assess the technical quality, feasibility, and overall contribution of this Final Year Project. Moreover, this paper could be a great reference for coming students or researchers dealing with assistive technologies, computer vision, or AI-powered navigation systems. Tech-lovers and developers concerned about accessibility issues will profit from the comprehensive description of methodology, design choices, and experiments discussed in this document as they will be able to build upon it for more innovation. Besides the educational institutions, the concepts and the suggested solution will be of great value to the organizations and NGOs that aid visually impaired people as a possible future focus for their attention and possible implementation and deployment on larger scale.

1.3 Definitions, Acronyms, and Abbreviations

Following is the list of all important definitions, acronyms, and abbreviations used in this document:

FYP: Final Year Project

AI: Artificial Intelligence

YOLO: You Only Look Once - An image classification Model

UI: User Interface

UX: User Experience

API: Application Programming Interface

CNN: Convolutional Neural Network

DNN: Deep Neural Network

BiLSTM: Bidirectional Long Short-Term Memory

SSAE: Sparse Stacked Autoencoder

VLM: Vision-Language Model

MR: Mixed Reality

TTS: Text-to-Speech

ETA: Electronic Travel Aid

EOA: Electronic Orientation Aid

PLD: Position Locator Device

LiDAR: Light Detection and Ranging

IoT: Internet of Things

HAR: Human Activity Recognition

OCR: Optical Character Recognition

SDG: Sustainable Development Goal

GPS: Global Positioning System

COCO: Common Objects in Context (image dataset)

UR: University of Rzeszów (Fall Detection Dataset)

Open Images: Open-source annotated image dataset

ABF: Adaptive Bilateral Filtering

GhostNet: Ghost Convolutional Neural Network (lightweight CNN backbone)

BiFPN: Bidirectional Feature Pyramid Network

VEO: Visual Environment Optimization

PRISMA: Preferred Reporting Items for Systematic Reviews and Meta-Analyses

MR.NAVI: Mixed-Reality Navigation System

VocalEyes: Vision-Language-based assistive navigation model

EDLES-SIAM: Ensemble of Deep Learning for Enhanced Safety in Smart Indoor Activity Monitoring

RGB-D: Red Green Blue + Depth (Camera)

SSD: Single Shot Detector (object-detection algorithm)

GPU: Graphics Processing Unit

CPU: Central Processing Unit

MLP: Multi-Layer Perceptron

SVM: Support Vector Machine

WCAG: Web Content Accessibility Guidelines

1.4 Conclusion

In conclusion, this introductory chapter established the motivation and importance of developing an AI-powered navigation system for visually impaired individuals. It outlined the purpose of the document, the intended audience, and the key terminologies used throughout the report. The remainder of this report is structured as follows: Chapter 2 presents the project vision, problem statement, goals, and scope. Chapter 3 provides a detailed literature review and competitor analysis. Chapter 4 describes the software requirements specification, including functional and non-functional requirements. Finally, chapter 5 outlines the proposed approach and methodology adopted for the project. Chapter 6 presents the high-level and low-level system design, including architectural strategies and policies.

Chapter 2 Project Vision

This chapter presents the vision of the proposed project **Sighteo: Navigate by Sight**. It defines the problem domain, elaborates on the challenges faced by visually impaired individuals, and provides a formal problem statement. Furthermore, the chapter outlines the goals, objectives, scope, and constraints of the system. It also highlights the Sustainable Development Goals (SDGs) addressed by this project and describes the key stakeholders and their roles. The aim of this chapter is to clearly establish the foundation upon which the technical design and implementation of the system are built.

2.1 Problem Domain Overview

The project is a mobile app that allows visually impaired individuals to navigate indoors safely and independently. The application uses the real-time camera feed from the phone with on-device object detection models to process the visual information and give haptic and/or audio cues immediately. Unlike other systems that give high-level spatial awareness and demand additional wearable devices, this solution focuses on obstacle-free movement and directional guidance indoors (rooms, corridors, offices, malls). The system is intended to be developed with characteristics like low latency, high accuracy, and a basic, user-friendly interface so that a visually impaired person can get fast and clear directions for avoiding obstacles and choosing a safe route.

2.2 Problem Statement

Visually impaired people frequently encounter safety and mobility challenges indoors: static furniture, low hanging obstacles, dynamic obstacles (people, carts), and clutter that a white cane or guide dog may miss at head or torso level. Existing approaches either rely on mapped environments, additional hardware, volunteer assistance, or provide only object identification and distance, not continuous, actionable navigation guidance tailored for indoor walking. The problem addressed here is: **How to provide fast, reliable, and accessible indoor obstacle detection and guidance using a mobile device camera and sensors, so visually impaired users can navigate common indoor settings independently and safely.**

2.3 Problem Elaboration

The navigation problem for visually impaired users decomposes into several sub-problems:

- **Real-time Obstacle Detection:** Detect common obstacles (chairs, tables, doors, stairs, people, boxes) from a monocular camera stream at walking speed. Detection must work under variable

- **Distance Estimation and Prioritization:** Estimate the relative distance and trajectory of detected obstacles (static and moving) and prioritize which obstacles present immediate collision risk.
- **Direction and Path Guidance:** Convert detection and distance into short, clear guidance that indicates a safe direction to move. indoor lighting and clutter.
- **Feedback Design (Audio and Haptic):** Create a feedback layer with concise text to speech prompts, and patterned haptic cues for immediate reactions.
- **Performance and Resource Constraints:** Allow the application to run on an average mobile device by keeping low CPU/GPU, memory, and battery usage.
- **Usability and Safety Testing:** Conduct testing with end users to minimize confusion in interpreting feedback cues. Build safety fallbacks such as immediate stop cue on sudden appearnce of a highly close object.

2.4 Goals and Objectives

Through Sighteo, we aim to make navigation easier and affordable for visually impaired individuals. This section covers a breakdown of the goals and objectives we aim to accomplish through this project.

- Deliver a reliable mobile-based indoor navigation aid that reduces collisions and improves independent mobility for visually impaired users.
- Optimize the system for indoor scenarios so detection and guidance are accurate and low-latency.
- Implement and evaluate an on-device object detection pipeline that achieves real-time frame rates for walking scenarios.
- Add distance or proximity estimation and a simple rule-based path planner that issues immediate directional instructions.
- Design and validate an audio and haptic feedback scheme based on user testing.
- Test the system in representative indoor environments (classrooms, corridors, shopping aisles, living rooms) and measure collisions, false positives, latency, and user satisfaction.

2.5 Project Scope

The project intends to create a mobile application for indoor navigation help aimed at visually impaired users in design, implementation, and evaluation. The application will:

- The primary input for the system will be the mobile's camera feed.

- Real-time object detection and proximity estimation will be done, and the results will be translated into audio and haptic guidance.
- Support only indoor environments (offices, homes, malls, corridors). Outdoor navigation, GPS-based routing, public-transport integration, and pre-mapped building routing are out of scope.

This narrower scope allows focused optimization for indoor-specific challenges (lighting, clutter, head-/waist level obstacles) and user experience design that does not need to handle outdoor variability.

2.6 Sustainable Development Goal (SDG)

The section shall provide an overview of how Sighteo aligns with the SDGs to create meaningful social impact. This section identifies and describes the interconnectedness of our solution with global objectives seeking to enhance equity, accessibility, and community well-being. Below, each sub-section details why both SDG 10 and SDG 11 are critically associated with the goals and long-term vision of our project.

2.6.1 Reduced Inequalities

One of the Sustainable Development Goals addressed by our research is **SDG 10: Reduced Inequalities**. This goal emphasizes empowering marginalized groups and ensuring equal opportunities for all individuals. One of the major problems visually impaired persons experience is a lack of navigation solutions that are accessible. This leaves them unable to move independently which effects their social life negatively.

Our project is aligned with the SDG by introducing an assistive technology whose main characteristic is to offer visually impaired persons freedom and independence. Discrimination that impaired users face in their everyday lives is reduced by Sighteo's independant navigation support.

Figure 2.1: Reduced Inequalities

2.6.2 Sustainable Cities and Communities

Another relevant SDG is **SDG 11: Sustainable Cities and Communities**. It highlights the importance of making cities inclusive, safe, and accessible for all. A major challenge for visually impaired individuals in urban environments is navigating public spaces. Although Sighteo will be tuned more specifically for indoors environments but this does not restrict it from use in certain controlled outdoor spaces. Traditional aids like canes leave users vulnerable in larger spaces.

Our project that merges AI-driven object recognition with real-time feedback. This will contribute to making cities and communities that are more inclusive, resilient, and supportive of people with disabilities. This comes in harmony with the global mission of SDG 11.

Figure 2.2: Sustainable Cities and Communities

2.7 Constraints

Although promising, Sighteo still has some limitations. Those limitations are listed below:

- **Environmental Factors**: Low-light conditions, crowded environments, and presence of fast moving obstacles may cause variance in detection accuracy.
- **Dataset Limitations**: Specialized datasets for visually impaired navigation are highly limited. This may require us to adapt or create a custom dataset.
- **Testing Environment**: Initial deployment will focus on controlled or semi-controlled indoor spaces. This may be expanded to testing in complex outdoor environments in the future.
- **User-Centered Design Issues:**: Designing for people with visual impairments fundamentally relies on iterative feedback and usability testing with real users. That does not always become easily possible.

2.8 Business Opportunity

The global assistive technology market is expanding rapidly. This is due to rising awareness of inclusivity. Current solutions rely on costly hardware (e.g., LiDAR sensors) or require constant human intervention, which limits their accessibility and scalability.

Sighteo can be a game-changer by providing a low-cost AI-based navigation system which can be easily availed through the usage of a smartphone. The technology can be integrated in NGOs, hospitals, and smart city initiatives. The increased demand in low-cost and mass-scale assistive technology can transform the project into a business opportunity.

2.9 Stakeholders Description / User Characteristics

The following section identifies key stakeholders in Sighteo and outlines specific roles for each group in the development and impact of the project. We also highlight some goals and expectations of these stakeholders and show how their needs shape the direction of the system. We then give a summary of main problems to be addressed for each group by Sighteo.

2.9.1 Stakeholders Summary

- **Primary Users**: Visually impaired individuals trying to seek safe, independent navigation.
- **Secondary Users**: Caregivers, and families having visually impaired individuals.
- **Project Team**: Developers/AI Engineers responsible for tasks like model training, deployment, and application design. Testers for validating the use of the application.

2.9.2 Key High-Level Goals and Problems of Stakeholders

- **Visually Impaired Users**: Need a reliable real-time navigation support for reducing dependency on others. Their main challenge is safety. They will expect accuracy from the system to trust it.
- **Caregivers and Families**: Want tools that provide assisstance reducing their own involvement. Their concern may be affordability and ease of use.
- **Developers/Researchers**: Aim to deliver a robust on-device system. The system must be capable of handling real-time object detection. Their challenge is balancing accuracy, latency, and connectivity requirements.

2.10 Conclusion

This chapter introduced the motivation, scope, and direction of the Sighteo project: defining the problem that people with visual impairments face and framing a solution with respect to low cost, accessibility, and real-time indoor navigation. The problem domain was established, the objectives of the system were explained, and its societal impact was underlined in relation to global development goals. By identifying stakeholders, constraints, and opportunities, this chapter gives clear ground to the technical decisions and design strategies presented later in the text.

Chapter 3 Literature Review / Related Work

This chapter of our report documents past literary works and existing technologies in the domain of assistive navigation for visually impaired individuals. Through this literature review, we aim to identify trends, strengths, and limitations of existing methods and systems. Academic researches were analyzed alongside existing commercial applications to provide a view of the landscape of this project. For each literary work or application, there is a summary, critically analysis comparison with Sighteo. The chapter concludes with summary tables that precisely mentions the findings from each work discussed.

3.1 Definitions, Acronyms and Abbreviations

Some of the important definitions, acronyms, and abbreviations are given below:

CNN: CNN stands for Convolutional Neural Network. It is a class of deep neural networks widely used for image recognition and classification.

VLM: VLM stands for Vision Language Model. It is an AI model that combines visual and textual data for understanding of context.

MR: MR stands for Mixed Reality. It is a technology that blends real and virtual environments. It is often used in advanced assistive navigation systems.

TTS: TTS stands for ”Text to Speech”. It is a library that converts text into audio output.

ETA: ETA stands for Electronic Travel Aid. It can be any device that provides information to assist visually impaired users in mobility.

EOA: EOA stands for Electronic Orientation Aid. These are systems that support orientation by providing information about the environment.

PLD: PLD stands for Position Locator Device. It is a device used to track the position of the user during navigation.

LiDAR: LiDAR stands for Light Detection and Ranging. It is a sensor that measures distances using laser by creating three dimensional maps.

COCO: COCO stands for Common Objects in Context. It is a large scale dataset used for object detection tasks.

ABF: ABF stands for Adaptive Bilateral Filtering. It is a technique used to preserve image edges used in pre-processing for Deep Learning Models.

BiLSTM: BiLSTM stands for Bidirectional Long Short-Term Memory. BiLSTM is a structure which feeds the sequence of the available data in forward and backward passes. It perceives the contextual dependencies.

SSAE: SSAE refers to the Sparse Stacked Autoencoder. It is the deep structure in the neural networks

utilized for unsupervised feature learning. It also allows dimension reduction.

DNN: DNN stands for the phrase Deep Neural Network. It's a multi-hidden layer network. It's extensively utilized while working with complex relations and data patterns.

IoT: IoT represents the Internet of Things. It refers to the collection of interconnected devices and sensors that communicate and exchange information in order to benefit from increased system intelligence.

UR Fall Detection Model: UR Fall Detection Model means University of Rzeszów w Fall Detection Model. It can be used for the evaluation of human activity recognition systems.

PRISMA: PRISMA refers to the Preferred Reporting Items for Systematic Reviews and Meta-Analyses. It includes the guidelines for conducting systematic reviews. It also allows for the transparent reporting of the studies.

GhostNet: The GhostNet stands for Ghost Convolutional Neural Network. It's a light-weighted CNN and produces feature maps. The maps assist in minimizing computations.

BiFPN: BiFPN stands for Bidirectional Feature Pyramid Network. It is a feature aggregation module. It is used in object detection.

VEO: VEO stands for Visual Environment Optimization. It is a framework that enhances node selection and path optimization in AI based navigation systems.

OCR: OCR stands for Optical Character Recognition. This computer vision technique extracts text from images and videos.

MR.NAVI: MR.NAVI stands for Mixed Reality Navigation System. It is a platform that integrates MR technology with spatial audio and visual cues.

HAR: HAR stands for Human Activity Recognition. It is the automated classification of human actions like walking or falling from sensor or vision data.

Open Images: Open Images is a dataset of annotated images used for object detection and classification.

3.2 Detailed Literature Review

This section of the report documents the review of literature consisting of past researches, studies, and existing competing solutions related to assistive navigation for visually impaired individuals. It discusses studies that introduce smartphone-based systems, object recognition applications, and real-time pathfinding algorithms, analyzing their strengths and limitations. By examining prior research and market alternatives, this review establishes the foundation for our proposed system, identifies critical gaps, and positions our work as a step forward in AI-powered navigation assistance.

3.2.1 A Smartphone-Based Mobility Assistant Using Depth Imaging for Visually Impaired and Blind

This work is one of the first efforts to investigate how smartphone depth sensors can support the mobility of visually impaired and blind users. This study focuses on demonstrating that depth information captured directly from a phone will be able to detect nearby obstacles without using any external devices. By leveraging the integrated sensing capabilities of modern smartphones, this system can offer basic spatial awareness and timely obstacle alerts. On the whole, the work illustrates that smartphone-based depth imaging is effective for serving as a practical basis in assistive navigation.

3.2.1.1 Summary of the Article

See, Sasing, and Advincula [**?**] introduced a smartphone-based mobility assistant for visually impaired and blind individuals that utilizes depth imaging technology. The core of their system lies in leveraging the built-in depth cameras of modern smartphones to provide users with spatial awareness and obstacle detection without the need for additional hardware. The technology extracts depth data from 23 coordinate points that are located within the scene that is being captured and then it divides the obstacles into the prescribed zones: head zone, torso zone, ground zone, or full body obstacles. The maximum range for obstacle detection mentioned was 1.6 meters which is the range suitable for being aware of the objects around.

Furthermore, the study not only indicated the possibility of having obstacle-detecting technology for blind people but also unveiled a new feature that helps directing visually impaired people to the areas where they want to look for things. The system applies various kinds of feedback very skillfully by offering the users both sounds and vibrations thus making it appropriate for a wide range of users' preferences. The process design has integrated voice and gesture controls, thus making the interface more versatile and user-friendly with more than 80% success rate confirming the effectiveness of the proposed method.

3.2.1.2 Critical Analysis of the Research Item

The study makes significant contributions to the field by examining depth imaging as a means to improve assistive navigation systems. To begin with, the most significant disadvantage of the study is the limited detection range of 1.6 meters, which might be insufficient. Additionally, the system used only 23 fixed coordinate points for collecting depth data, which was a problem from the perspective of reliability. Even though less computationally intensive, this approach might miss small or strangely shaped obstacles. Moreover, while the system alerts the users to the presence of the obstacles, it does not provide them with

the means of direction or map planning. The users will have to take navigation decisions themselves, which could still be very difficult in case they are not familiar with the area. Moreover, the visually impaired find the use of voice and gesture inputs as an extra cognitive load. A person with impaired vision may find it hard to give a voice command or perform a gesture to change the system settings. Static object detection and obstacle alerts rather than continuous navigation in dynamic environments, is the main focus of the research. Changes in lighting, reflective surfaces, or multiple moving obstacles are issues which were not deeply analyzed, leaving the robustness of the solution area of concern under real-world conditions.

3.2.1.3 Relationship to the Proposed Research Work

The most intuitive way of providing feedback is revealed by the new methodology of categorizing hindrances as particular body zones. The limitations of See et al.'s [**?**] system highlight the areas that our project aims to address. Their work is meant for indoor and outdoor settings, but it fails to provide the needed assistance particularly in indoor environments where the challenges are more serious due to narrow hallways, irregular furniture arrangements, and cluttered obstacles. By contrast, our project focuses exclusively on indoor spaces. Unlike See et al.'s work, which primarily alerts users to obstacles, our proposed system aims to provide real-time guidance by indicating safer pathways and suggesting navigation directions. This transforms the system from a passive warning tool into an active guide. Additionally, while their system uses a fixed set of coordinate points for detection, our project intends to employ more comprehensive real-time object detection techniques to improve accuracy in identifying both static and dynamic obstacles. Lastly, See et al. [**?**] set a good platform by proving that depth imaging on smartphones can benefit visually impaired users. But the restricted range of detection, lack of directional cues, utilization of fixed points, and focus on multimodal control all indicate vasts areas of improvement. Our research builds on these observations in the direct sense of being centered on indoor spaces and providing better navigation assistance.

3.2.2 Android-Based Object Recognition Application for Visually Impaired Users

This work presents a simple yet meaningful attempt to develop an Android-based object recognition app supporting the needs of visually impaired users. The authors focus on showing how lightweight models can be deployed directly on the mobile device, independent of cloud services, to make assistive tools more accessible and affordable. By combining real-time object detection with a speaking interface, this study demonstrates a practical method for audio cues about environmental understanding by users. It demonstrates how on-device AI can improve everyday independence in visually impaired users.

3.2.2.1 Summary of the Article

Bhagwat et al. [**?**] introduce an Android app in this research that is intended to assist visually challenged users in comprehending their environment. The app detects objects in real time. It grabs the live video frames from the camera of the smartphone. Utilizing TensorFlow's Object Detection API, it identifies objects on the scene. The model is transformed into TensorFlow Lite. This enables offline, device-local processing, eliminating reliance on cloud or network availability. Once an object is recognized, it is converted into an audio cue using the device's Text to Speech (TTS) engine. The authors claimed that about 90 percent accuracy was achieved by the system in object detection in their test cases. The paper also describes a component for real-time text reading (recognizing printed text in the live scene) alongside object recognition, although the primary focus is object detection and voice feedback.

3.2.2.2 Critical Analysis of the Research Items

In this paper, the authors show a practical method for implementing object detection in a mobile app that has offline functionalities. Using the SSD architecture is a great decision for resource-limited environments. The choice of Tensorflow Lite is an indicator that the developers are aware of the issues related to the practical deployment of mobile applications. However, the limitations of the system in terms of design, evaluation, and claims become clear on careful examination. Firstly, the 90% accuracy they claim is not only about false positives, latency, or performance in difficult indoor conditions like low light, occlusion, or clutter. It mainly concerns either static scenes. Secondly, the system only informs the user about the names of the objects only. It does not offer any guidance for navigation. Thirdly, the system ignores dynamic obstacles such as moving people. Fourthly, there is not any mention of real-time performance such as frames per second or latency, or energy consumption, which are all critical for any assistive tool. In addition, the ability to read texts is a nice feature to have, but the authors fail to mention how dependable this is when lighting changes, the text is placed at an angle or on a curved surface, and how the user experience is impacted by switching between recognition of objects and reading of texts. The usability of the system under walking real-life conditions in rooms where a lot of objects or surfaces are mixed is not completely tested and reported.

3.2.2.3 Relationship to the Proposed Research Work

Bhagway et al.'s [**?**] work offers a useful reference point and a technological baseline. It demonstrates the feasibility of running object detection and voice feedback in real time on a mobile device. It establishes the fact that mobile phones can handle object recognition without relying on server infrastructure. However, the gaps in their work align closely with the problems Sighteo aims to solve. Their system stops at object identification and vocal naming. It does not go further into active navigation guidance.

And, that is central to our goal. Because Bhagway et al. do not consider indoor specific constraints such as tight hallways, clutter, floor level hazards, or moving obstacles, their design is generic. Sighteo will build on this foundation. Through Sighteo, we aim to enhance performance providing continuous directional guidance. We will also focus on minimising latency, which was not the focus in this paper. After establishing that object recognition on a phone for the visually impaired is indeed possible, the paper leaves an open challenge of converting that recognition into safe real time navigation.

3.2.3 Efficient Real-Time Pathfinding for Visually Impaired Individuals

This work redirects focus from mere obstacle detection to the computation of safe real-time paths for visually impaired users. Instead of merely attaining awareness, it investigates how algorithms can lead users through dynamic scenes by computing collision-free paths as they move. The work emphasizes low latency, adaptability, and computational efficiency, demonstrating that effective pathfinding forms a backbone on which to base live navigation support. Overall, it shows how computation of real-time routes can enhance mobility beyond object detection alone.

3.2.3.1 Summary of Research Item

Ghahremanians and Mohammadi [**?**] present an approach to efficient real-time pathfinding designed for visually impaired individuals. Their work focuses on computing safe, collision-free paths as users move through environments with obstacles and possibly dynamic elements. The proposed method aims to reconcile several challenges: low latency, computational efficiency, and adaptability to changing surroundings. The algorithm is designed for real-time execution and therefore more applicable to live guidance, but less so for offline route planning. Despite the paper's focus being generic navigation, its path planning techniques in constrained environments are more applicable to assistive systems.

3.2.3.2 Critical Analysis of the Research Item

This work is worthwhile since it addresses the essential problem translating obstacle information into navigable routes, instead of merely detecting objects. Previous systems often get only up to the point of "alert user to object," while in Sighteo the problem solved is determining what path the person should move along in order to prevent collisions. Stressing real-time capability suggests the authors are aware of the importance of responsiveness, which in a navigation aid can be paramount. Negative aspects are that the publicly available summaries are uninformative about precise evaluation outcomes, particularly under indoor, cluttered, or dynamic conditions. It's also indeterminate how uncertain or noise-laden input (e.g. from a moving-camera phone's camera) is handled. Another drawback is whether the technique presumes a mapped environment (or pre-known layout), versus operating in arbitrary, unseen spaces.

Furthermore, the paper apparently targets quite widely (for general visually impaired navigation); we are not clear that it's particularly optimized for indoor limitations such as narrow hallways, objects at the person's level, or overlap furniture. The computational efficiency claims are promising, but we are uninformed about memory usage, energy usage, or how well it tolerates latency in low-spec devices (e.g. phones). Lastly, dealing with dynamic objects (moving people or moving objects) represents a serious practical problem, and it's indeterminate from the summary how stable their method is in this respect.

3.2.3.3 Relationship to the Proposed Research Work

This research in specific is up-to-date since it fills the gap in the literature from obstacle detection up to path planning. For our potential app, their research illuminates methodically how to compute in real time safe paths from sensor readings. It demonstrates that pathfinding can be a substantial part of a support for navigation aid and not merely a theoretical construct for research's sake. Meanwhile, our solution possesses a great number of features which are specific only to ours. We consider only indoor navigation with all its specific characteristics like restricted space, clutter, varying illumination, moving objects, shadowing effects, stairs, and narrow corridors. The approach described by Ghahremanians and Mohammadi [**?**] does not limit its applicability for indoor-only use and thus potentially overlooks some indoor-specific issues. Additionally, our application is intended to provide unobstructed guidance (directions, avoidance maneuvers) during user movement, perfectly synchronizing with user detection rather than relying on prior mapping or grand outdoor-style planning. Their approach is based on efficiency; thus, we are able to employ their strategies for latency reduction or simplified path planning but we need to make sure that they are adapted to effectively work in the challenging situations created by noisy, occluded, and rapidly changing indoor mobile camera input. In short, their research informs the path planning component of our system, but we must extend it with indoor specialization, tighter integration with object detection, and user-centric feedback (audio + haptic) to make it work in practice in indoor walking assistance.

3.2.4 Improved YOLOv5 Algorithm Combined with Depth Camera and Embedded System for Blind Indoor Visual Assistance

This work investigates the improvement of YOLOv5 by fusing depth camera data, optimized embedded hardware, and refinements in the model architecture for indoor assistance applications. The authors investigate how depth information may strengthen detection performance, especially when the scenes are cluttered or objects are very close to the camera. The paper also underlines how architectural adjustments will lower the computational load without performance degradation and make the system more apt for real-time processing on lightweight devices used indoors.

3.2.4.1 Summary of the Research Item

Zhang et al. [**?**] propose an enhanced YOLOv5 algorithm integrated with a RealSense D435i depth camera and a Raspberry Pi 4B to create an indoor assistive system for visually impaired individuals. Their approach replaces the YOLOv5 backbone with GhostNet to reduce parameters, incorporates a coordinate attention mechanism for better feature representation, and uses a bidirectional feature pyramid network (BiFPN) for improved feature fusion. In comparison with the baseline YOLOv5s models, the upgraded version reduces the size by 42.4% and the number of parameters by 47.9% while improving the recall by 1.2% and keeping the precision stable. The system also accommodates voice input and yields audio feedback with object detection and ranging, making the system more suitable for use in everyday indoor applications among the visually-impaired.

3.2.4.2 Critical Analysis of the Research Item

One of the key advantages of the work is the good tradeoff between detection accuracy and computational efficiency such that the system can be run on handheld devices such as Raspberry Pi. The combination of the GhostNet and the BiFPN shows good architectural innovation in attaining light-weight design and stable performance. The incorporation of a depth camera also corrects one key weakness in the literature on 2D vision systems, which is the inability in the state-of-the-art literature to estimate distances accurately. Despite the foregoing advantages, the system does have some key weaknesses. Its training dataset is quite small and comprised only of some indoor objects (e.g., bottle, cup, key), so generalization is limited. In addition, low-light and speech accent variations cause concerns in the usability in the real world. Even though the work was conducted in highly restricted indoor conditions, questions remain about its scalability and robustness in more varied and more complicated conditions.

3.2.4.3 Relationship to the Proposed Research Work

This work immediately relates with our project concept as it validates the possibility of combining embedded devices and computer vision in the assistive navigation task. Even though our system utilizes the smartphone camera, and not a depth camera, the architectural optimizations proposed in Zhang et al. [**?**] provide potential directions in the tradeoff between computational load and precision. In its concern with fusion of auditory feedback and real-time object detection, its orientation also strongly corresponds with our aim in the provision of intuitive direction-finding assistance to the visually impaired.

3.2.5 Assistive Navigation Technologies: A Comprehensive Overview

The following work provides a general overview of assistive navigation technologies developed for visually impaired users. In this work, a wide range of existing solutions is reviewed; also, there is a categorized overview by sensing methods, system architectures, and techniques for user interaction. This paper maps out how different sensors and feedback approaches are used across current systems, giving a clearer understanding of the technological landscape. This broader perspective brings to light not only the strengths of the available methods but also the points that still require attention in future designs.

3.2.5.1 Summary of the Research Item

Abidi et al. [**?**], in this paper, reviewed works by scientists who experimented with a range of devices to assist the visually impaired. These include wearable such as haptic vests and smart glasses, and non-wearable ones such as navigation sticks and sensory systems. Both the RGB-D cameras and tactile sensors can be utilized so that the user can have a greater understanding of his/her surroundings, whereas IoT-powered solutions and satellite guidance allow more users to access the system simultaneously [**?**]. Furthermore, the study reveals that the human centered design approach is the crucial factor for the usability and customization and thus it will resultin an increase in the quality of life of the blind and visually impaired individuals [**?**].

3.2.5.2 Critical Analysis of the Research Item

One of the best strengths of this piece is its thorough categorization of current solutions and discourse of their technical underpinnings, which provides a strong foundation for future research. However, thepaper largely canvasses existing technologies without critically confronting the limitations of real-worlddeployment, including internet access, expense, and obstacles to user adoption. Such gaps mean thatwhereas the review is complete, it is more descriptive than evaluative regarding practical application.

3.2.5.3 Relationship to the Proposed Research Work

For our proposed work, this review is a fine benchmark by identifying the importance of integrating artificial intelligence and mobile-based solutions for visually impaired navigation. Since our project is designed to apply a smartphone camera for displaying real-time navigation directions, the outcomes directly support the relevance and feasibility of our approach while also reinforcing the need to address practical constraints like connectivity reliability, which are hardly addressed in the review [**?**].

3.2.6 Assistive Systems for Visually Impaired Persons: Challenges and Opportunities

This review covers the key challenges intrinsic to the design of effective assistive systems and emphasizes how new sensing technologies and AI methods offer possibilities for their improvement. It underlines issues such as accuracy, reliability, user comfort, and real-world functioning. It also discusses opportunities opened up by both advancements in mobile hardware and lightweight machine learning models. In general, this review outlines core factors that will eventually shape usability and influence whether such technologies are likely to be adopted by visually impaired users.

3.2.6.1 Summary of the Research Item

Okolo et al. [**?**] offer a review targeting the challenges and opportunities in developing assistive systems for the visually impaired, with specific interest in navigation aids. The paper categorizes assistive technologies into sensing inputs (e.g., smartphones, RFID, depth cameras, infrared, ultrasonic, RGB-D), feedback mechanisms (audio, tactile, vibration, haptic), and IoT-based systems. It highlights the need for the incorporation of artificial intelligence, computer vision, and mobile technologies to facilitate the improvement in obstacle detection, spatial understanding, and real-time navigation assistance. The review also covers both commercially used tools (e.g., SmartCane, WeWalk) and research prototypes in order to get a wide view of existing technology developments.

3.2.6.2 Critical Analysis of the Research Item

One of the strengths of this work lies in its in-depth categorization of technologies and comparative analysis of AI- and IoT-based solutions. It extends merely describing tools by including limitations such as bulkiness, expense, low performance in well lit conditions, and difficulties in user uptake. Nevertheless, the review mainly exists in the realm of theory and does lack empirical analysis of system usability in real-world settings. Furthermore, whereas it lays out future suggestions such as multi-modal feedback and mobility, the absence of concrete design guidelines or user-centric confirmation stands out as a significant omission.

3.2.6.3 Relationship to the Proposed Research Work

This paper also links well with our project idea because it points out the necessity for mobile and AI-based navigation systems. Since our system relies on a smartphone camera for obstacle detection and directions, the result in terms of limitations in vision-based systems (e.g., ineffectiveness in depth estimation in normal cameras) directly corresponds with the expected technical challenges we must transcend. Furthermore, the recommendation for light, userfriendly, and portable systems corresponds well with

our goal of developing a feasible, smartphone-based navigation system specifically with the visually impaired in view [**?**].

3.2.7 Technological Advancements in Human Navigation for the Visually Impaired: A Systematic Review

This systematic review details recent advances in navigation technologies designed for the visually impaired, putting current development into a larger technological and practical context. This review reveals emergent trends in sensing, localization, and feedback mechanisms, while also mapping the common design patterns that many systems adhere to. Simultaneously, this review identifies key gaps that still hold back widespread adoption due to usability, cost, reliability, or real-world performance issues. These insights, taken together, help clarify where future research and development should be focused.

3.2.7.1 Summary of the Research Item

Casanova, Guffanti and Hidalgo [**?**] provide a systematic review of 58 recent works from 2019 to 2024 on assistive navigation technologies for the blind and visually impaired. The authors emphasize that while advanced sensors and artificial intelligence driven systems are growing in importance, the current state remains such that people with blindness and visual impairments are far from fully benefited in terms of access and inclusivity. The review synthesizes the major innovations such as the integration of high-precision GPS with ultrasonic sensors, Bluetooth and haptic feedback systems, and deep learning-based algorithms for obstacle avoidance and route planning. The review lists the range from the simplest screen reading and navigation app assistance to the latest smart cane, smart shoe, smart belt and LiDAR-based technologies. Some examples are BlindRouteVision, which integrates Android apps with GPS and ultrasonic sensors, and wearable systems based on Raspberry Pi technologies that implement fuzzy logic and vibrotactile feedback for real-time safe navigation.

3.2.7.2 Critical Analysis of the Research Item

Among its key strengths, the review stands with its systematic and exhaustive coverage, employing PR the Accept Dismiss ISMA methodology (a protocol for enhancing the clarity and quality of reports of systematic reviews and meta-analyses) for ensuring rigor. It classifies solutions into smartphone-based systems, haptic interfaces, and AI-based naviagation aids and reports that 79% of the studied surveys were included with experimental validation, with haptic feedback having the highest prevalence as a modality. The review also pinpoints the weaknesses in existing systems, including dependency on GPS, high expense, and low uptake among users due to usability issues. The weaknesses suggest the directions towards more affordable, accessible, and user-oriented designs.

3.2.7.3 Relationship to the Proposed Research Work

This project directly substantiates our project's mission. It validates our decision to combine AI-powered object detection with audio and haptic feedback and sheds light on the gaps we aim to close, notably improving indoor navigation and designing with affordability and access in mind. We are building on the literature's strengths and aiming to avoid the weaknesses in order to create a useful solution.

3.2.8 Assistive Systems for Visually Impaired Persons: Challenges and Opportunities

This section looks into the various machine learning systems developed for visually and hearing-impaired people. It underlines how new models contribute to recognition and perception and, in general, to accessibility. Much importance is given to the way these technologies interpret an environment surrounding real life, classify sounds, and assist in daily navigation, with an increased role being given to deep learning for accuracy and responsiveness.

3.2.8.1 Summary of the Research Item

Patel et al. [**?**] give an overview of machine learning methodologies used to aid visually and hearing-impaired people, including classic ML and deep learning. For hearing, they give an overview of models like SVM, Random Forests, and MLPs for speech and sound classification tasks. For vision, they offer a rundown of real-time object detection models like YOLO, SSD, and RetinaNet that enable scene understanding and navigational help on wearables and smartphones. The article also cites Generative AI applications and points out how fine-tuning and personalization are typically needed to fit user needs. Generally, the paper follows the trajectory from device, dataset, and algorithm to referring to the ways in which these approaches become useful tools for assistance.

3.2.8.2 Critical Analysis of the Research Item

Strength lies in the fact that the paper covers two sensory modalities and clearly separates method by task and model family, making it a good place to start design selection. It is also concerned with real-time constraints and the importance of model-specific user tuning, which brings one close to real-world in deployment. Being a general survey, it does not include head-to-head comparisons under a common protocol, and therefore comparative performance has to be deduced from a number of studies. On-device limitations (battery, thermal throttling) and user study discussion are minimal, and there are unresolved issues of long-term acceptability and cost in field use. These are reasonable trade-offs for a survey but have to be recognized when progressing insights to an operational system.

3.2.8.3 Relationship to the Proposed Research Work

For Sighteo's indoor navigation emphasis, the paper reaffirms employing contemporary object detectors (e.g., YOLO/SSD/RetinaNet) and selecting an appropriate dataset to find accuracy vs. speed balance. It also encourages us to think through personalization and feedback modalities upfront, as most assistive solutions take advantage of user-adjusted thresholds and interfaces. Practically, this review helps us shortlist model families for experimentation, set evaluation criteria around latency and reliability, and plan for future extensions (for example, exploring lightweight Generative AI features for descriptions or captions once core navigation is stable).

3.2.9 Ensemble of Deep Learning and IoT Technologies for Improved Safety in Smart Indoor Activity Monitoring for Visually Impaired Individuals

A framework of deep learning and IoT-based indoor activity monitoring is introduced in this literature. Although the work is not focused on navigation, it still provides insight into an ensemble model, sensor fusion, to improve the safety and situational awareness of a visually impaired user. This perspective broadens our view of how multiple streams can work in tandem to create more reliable assistive environments.

3.2.9.1 Summary of the Research Item

Al Duhayyim [**?**] proposed an ensemble-based deep learning and Internet of Things (IoT) framework known as EDLES-SIAM (Ensemble of Deep Learning for Enhanced Safety in Smart Indoor Activity Monitoring) to enhance safety and independence for visually impaired individuals in indoor environments. The system combines multiple deep learning architectures, including Deep Neural Networks (DNN), Bidirectional Long Short-Term Memory (BiLSTM), and Sparse Stacked Autoencoders (SSAE), into a unified ensemble to perform accurate human activity recognition (HAR). The approach starts with Adaptive Bilateral Filtering (ABF) to filter sensor and image noise and then feature extraction is done using ResNet50 for spatial feature extraction. The ensemble outputs are summed up sequentially to infer human activity as well as abnormal activity like falls or unsafe movement. UR Fall Detection dataset was experimented on test and classification accuracy because of it 99.25% in comparison to standalone deep models such as VGG16, VGG19, and certain ResNet versions. The paper discusses how combined IoT and deep learning systems individually contribute effectively towards monitoring visually impaired people in real-world scenarios where assistance is essential.

3.2.9.2 Critical Analysis of the Research Item

The robustness of this work lies in its hybrid nature, borrowing the advantages of recurrent and autoencoder-based temporal analysis and CNN-based feature extraction. Its use with real-time sensing using IoT devices enhances over static image processing to offer real-time situational awareness. Adaptive Bilateral Filtering (ABF) usage also enhances image quality, leading to increased recognition accuracy. Nevertheless, the model's high computational cost restricts its portability to low-capacity platforms like smartphones. Relying on a single dataset (UR Fall Detection) in an effort to reduce generalization to more varied environments is another restriction. In addition, the lack of user trials involving visually impaired users ensures that real-world usability, latency, and energy efficiency of the system are not realized. Although the findings are encouraging for smart indoor safety, accessibility and scalability continue to be key challenges.

3.2.9.3 Relationship to the Proposed Research Work

Al Duhayyim's [**?**] work is of direct applicability to the suggested smartphone-based navigation system, as it illustrates how ensemble deep learning may enhance the perception model's robustness used for assistive safety. The idea of combining deep learning with IoT-based sensing supports the significance of multi-modal data in improving situational awareness. Yet, in contrast to EDLES-SIAM based on high-performance IoT configurations and ensemble computation, the suggested project seeks to provide real-time perception based on only a martphone camera and a light AI model compiled for mobile. It ensures affordability, mobility, and freedom from external sensors or high-end edge hardware. Observations from this work affirm the worth of preprocessing, feature integration, and ensemble learning concepts but call attention to the need to streamline these processes for low-power, real-time application applicable to daily use by the blind.

3.2.10 VocalEyes: Enhancing Environmental Perception for the Visually Impaired through Vision-Language Models and Distance-Aware Object Detection

This work introduces an assistive system that incorporates vision-language models into a distance-aware detection pipeline. It demonstrates how spatial information significantly enhances contextual understanding and increases the environmental awareness of the user. By offering more descriptive and meaningful feedback, this departs from the simple detection of objects and provides a richer interpretation of the surroundings, one that's particularly valuable for assistive navigation.

3.2.10.1 Summary of Research Item

Chavan et al. [?] propose VocalEyes, a real-time assistive system designed to improve situational awareness for visually impaired individuals. The system uses a quantized and fine-tuned Florence-2 large vision-language model, optimized for 4-bit accuracy, to process the live video input and make the deployment on low-power devices like the NVIDIA Jetson Orin Nano. With a latency of 5 frames, the system is capable of giving contextual descriptions of objects, pedestrians, and obstacles, distance estimation included. For feedback purposes, it has incorporated Parler TTS Mini, a compact text-to-speech engine that supports 34 different speaker types and offers customizable voice styles, thus giving audio guidance that is user preference-based. The research considers the system's accuracy, efficiency, and usability and gives an impression of it being a real-time, low-cost navigation support solution.

3.2.10.2 Critical Analysis of the Research Item

A major strength of this work is its successful adaptation of a large model (Florence-2) for edge deployment. The system does fine-tuning and quantization to achieve computational efficiency and contextual precision. Value is added beyond pointable object recognition in the form of the distance-aware detection and the personalized TTS with the possibility of customization. Even so, there are limitations. The requirement for NVIDIA Jetson hardware can limit accessibility with respect to smartphone-only implementations, and the system's latency of 5 frames, although comparative low, can potentially compromise responsiveness in rapidly changing scenes. In addition, the analysis focuses heavily on accuracy and efficiency but sheds limited light on the longer-term user studies and the robustness in varied real-world conditions (e.g., low illumination, high-population indoor scenes, outdoor variations).

3.2.10.3 Relationship to the Proposed Research Work

VocalEyes [?] is a major reference for our work since it illustrates the great potential of tuning large AI models to be used in real-time assistive applications. The system works on top of edge hardware but we are striving to provide similar features, namely contextual object detection and real-time feedback, with the help of a smartphone camera only, to make the system available at a larger scale. The use of distance measurement and adaptive voice feedback in VocalEyes is very much in line with the intent of offering easy and lovable navigation support to the user. On the other hand, the problems of hardware dependence and latency raised in the research pointed out that developing lightweight, mobile-optimized solutions that are independent of external processing devices becomes even more important, and this is exactly where our focus of research is.

3.2.11 MR.NAVI: Mixed-Reality Navigation Assistant for the Visually Impaired

This paper discusses using mixed-reality headsets for navigation. It combines spatial audio, object detection, and real-time guidance in a single wearable device and lets the user receive constant and intuitive support while he or she moves around complex environments. Blending virtual cues with real-world perception, it underlines how mixed reality has the potential to give visually impaired individuals a more confident and informed way to navigate.

3.2.11.1 Summary of Research Item

Pfitzer et al. [**?**] present MR.NAVI, a mixed-reality navigation system designed to enhance spatial awareness and mobility for visually impaired individuals. The system presents a complete navigation assistant by the combination of computer vision, natural language processing, and spatial audio feedback. Operated on the Microsoft HoloLens2, it integrates object detection, depth estimation, obstacle avoidance as well as public transit navigation under a single framework. MR.NAVI is operating at the same time as an Electronic Orientation Aid (EOA), Electronic Travel Aid (ETA), and Position Locator Device (PLD), filling the space that existing solutions often leave behind when they only provide one or two of these functionalities. The application provides audio and visual feedback of scene descriptions, collision avoidance guidance, and navigation directions, while voice commands are the main user interface for the users. User studies demonstrated that MR.NAVI improved participants' confidence in navigating unfamiliar environments, though limitations in spatial audio precision and environmental adaptability were noted.

3.2.11.2 Critical Analysis of the Research Item

The primary strength of MR.NAVI lies in its comprehensive integration of EOA, ETA, and PLD features into a single mixed-reality platform, which many competing systems fail to achieve. The utilization of multimodal feedback in the form of spatial audio, visual indication, and natural language response makes the system capable of use on a variety of visual impairments. Moreover, using available consumer hardware such as HoloLens2 shows pragmatism over very special-purpose custom-built devices. Yet, the dependency on HoloLens2 generates accessibility limitations since the device is expensive and less mobile than smartphone-based solutions. Dependence on high Wi-Fi connectivity and a server on the PC for processing-intensive tasks brings into question real-world usability and scalability. User studies, as enlightening as they were, used blindfolded non-sighted participants and not visually impaired ones, which compromises ecological validity. In addition, the outdoor performance of the system was troubled by lighting changes and sensor limitations, revealing the vulnerabilities of robustness in real-world settings.

3.2.11.3 Relationship to the Proposed Research Work

The findings of MR.NAVI [?] are significant to our target smartphone-based navigation system, Sighteo. Although MR.NAVI demonstrates the capabilities of integrating various assistive technologies in a single framework, its reliance on mixed-reality hardware makes it less practical for mass adoption. By contrast, our project aims at replicating the same functionalities with the least and simplest hardware consisting only of a smartphone camera, thus it is low-cost and accessible. The focus in MR.NAVI on natural feedback strategies (spatial audio and natural language) aligns with our goal to offer constant, natural guidance by audio and touch feedback. But while MR.NAVI was mostly tested with simulated environments and necessitated server support, our system will focus on light, mobile-optimized computation to support real-time guidance without the need for external devices. In essence, MR.NAVI demonstrates the advantages of multimodal, AI-driven navigation but also underscores the importance of portability, affordability, and independence from expensive hardware, principles central to our proposed work.

3.2.12 Visually Impaired Aid using Convolutional Neural Networks, Transfer Learning, and Particle Competition and Cooperation

This work presents a hybrid AI approach combining the strengths of convolutional neural networks, transfer learning, and a cooperative particle-based algorithm to enhance object classification performance. The authors blend these techniques in the pursuit of higher recognition accuracy for more reliable assistive support in users. This study shows how layered improvements in model design can translate into more dependable functionalities in the real world.

3.2.12.1 Summary of Research Item

Breve and Fischer [?] present a system that leverages convolutional neural networks (CNNs), transfer learning, and a novel particle competition and cooperation algorithm to support visually impaired individuals in navigation tasks. For the suggested method, the first step is to apply CNNs for environment object detection and classification, this ultimately leads to the identification of both the barriers and the related contextual cues. The transfer of knowledge from one field to another is also utilized in the form of a pre-trained model in deep learning thereby increasing the precision of the output. Besides this, the particle competition and cooperation strategy is another technique that positively influences the decision-making process regarding the object classification. The combination of the methods enables the system to produce more robust and accurate identification of environmental landmarks than individual CNNs. Their work demonstrates that the integration of multiple AI techniques can make a huge difference in assistive navigation systems.

3.2.12.2 Critical Analysis of the Research Item

The primary contribution of this research is the hybrid framework, combining CNNs and transfer learning with particle competition/cooperation in an effort to enhance classification accuracy. This combination addresses typical issues like small training sets and poor generalization in real environments. Transfer learning reduces the necessity of large budgets for training, hence making the solution more deployable. Nevertheless, the use of computationally expensive deep models within the system might restrict its portability and real-time suitability on devices like smartphones. The work focuses primarily on classification and detection of objects and does not include feedback information for how end-users are receiving feedback when they are navigating. The research is also not enabled with extensive real-world usability testing involving visually impaired participants, bringing its use in real-world environments with changing contexts into doubt.

3.2.12.3 Relationship to the Proposed Research Work

For our project, Breve and Fischer's article [**?**] is an informative resource regarding how transfer learning and CNNs can be used to improve object detection accuracy in assistive systems. Their approach shows that the power of several algorithms being able to increase performance can be harnessed, which also aligns with our goal of creating a trustworthy smartphone-based navigation assist. But whereas theirs is directly aimed at recognition accuracy, our solution builds on theirs by extending it with real-time navigation hints, collision detection, and multimodal feedback (auditory and haptic) for independent indoor mobility. By aiming at light, smartphone-based solutions, our system aims to circumvent the computational resource constraints of their solution while leveraging its advantage in AI-based object detection.

3.2.13 Enhancing Indoor Navigation for Visually Impaired Individuals with an AI Chatbot Utilizing VEO Optimized Nodes and Natural Language Processing

This work presents an interactive indoor navigation system based on an AI-driven chatbot, operating with natural language processing and a network of environment-optimized nodes to provide users with voice-based, step-by-step instructions. In this instance, the approach is being demonstrated by showing that familiar dialogue-based interactions may make navigation more intuitive and minimize the learning curve for those with a visual impairment.

3.2.13.1 Summary of the Research Item

Thandu and Rathinam [?] also introduced an AI indoor navigation system that leveraged Natural Language Processing (NLP) and VEO (Virtual Environment Optimization)-optimized nodes in order to guide visually challenged people through an interactive chat-like interface. The system automatically guides people in indoor spaces based on voice queries and gives directions in a conversational mode. The chatbot's engagement is with a cluster of VEO-optimized nodes that are technically affixed to the environment and thus present an avenue for dynamic path generation and guidance of users depending on the different contexts. The chatbot is empowered to discern user intent by means of NLP algorithms and subsequently offer feedback that is of great value and awareness to the situation. The entire approach is solely aimed at making indoor navigation more intuitive and interactive by means of dialogue-based support that in turn would lessen the barrier between accessibility and user engagement. The findings from the experimental study showed that this system not only had a very high response accuracy rate but also decreased disorientation of users when compared to the case of static voice guidance techniques.

3.2.13.2 Critical Analysis of the Research Item

The main merit of the work of Thandu and Rathinam is the hybridization of conversational AI with the already present indoor navigation in real-time. The system using NLP provides an apt and more human-like way for the visually impaired people to ask for directions as well as get guidance. The feature of having VEO-optimized nodes incorporated in the system allows the computing of the path to be done faster and the environment to be more adaptable. On the negative, the model only browses through communication and route response whereas it can also serve low-level perception tasks such as obstacle detection and environmental sensing. Due to this fact, the use of the device in such environments where visual feedback or real-time avoidance of hazards is a requirement becomes less suitable. Moreover, the researchers didn't state anything regarding latency, fault tolerance, or noise robustness performance indexes which are all essential for the assistive technologies. Another point indicating a tech reliance on mapping from preset nodes is the low scalability for bigger or unmapped indoor spaces.

3.2.13.3 Relationship of the Proposed Research Work

Thandu and Rathinam's work [?] is essential for Sighteo because it extends the area of assistive navigation to include AI-driven communication that is truly interactive. Although the Sighteo system under consideration only detects obstacles using the smartphone camera and creates spatial awareness, the augmentation of a conversational interface for guidance is a possible future move. The findings from the current research indicate that feedback systems that are adaptable, user-centered are essential in making navigation compliant. The Sighteo system not only accepts this but also proposes to replace

map-based navigation with real-time visual perception and to provide feedback in two forms (audio and haptic) thus offering both interactive and perceptual support without the need for an already defined environment.

3.2.14 Smart Assistive System for Visually Impaired People: Obstruction Avoidance through Object Detection and Classification

This article proposes an intelligent assistive system that incorporates object detection with ultrasonic sensing for safe navigation by visually impaired people. Merging the visual cues with accurate distance measurements, the system delivers timely audio warnings to users about obstacles in proximity. This layered sensing helps to make navigation more dependable and responsive.

3.2.14.1 Summary of the Research Item

Masud et al. [**?**] presented a new technological framework called smart assistive that would facilitate the visually challenged people's way through the environment by detecting and classifying the objects in their way. The amalgamation of computer vision and deep learning algorithms makes it possible to identify the nearby objects and classify them according to the risk level or distance in real-time. An ultrasonic sensor works alongside the stereo vision camera to get spatial information, which is then processed with a convolutional neural network (CNN) for the identification of the object. Upon detection of obstacles, the auditory feedback module notifies the user with spoken alerts about the type and distance of the obstacle. The system was tested in both indoor and outdoor scenarios and reached a high object recognition accuracy and a low latency rate, which resulted in the effectiveness of the system in terms of obstacle detection and collision prevention. The authors pointed out the great potential of using a combination of vision and sensor-based data for creating a robust assistive navigation system.

3.2.14.2 Critical Analysis of the Research Item

One of the main benefits that arose from the research conducted by Masud and his colleagues is the hybrid sensing technique based on vision and ultrasound data that boosts reliability in situations with rapidly changing light conditions or partial visibility. The synchronization of real-time object detection with audio feedback indicates that it is suitable as a working tool for the navigation in daily circumstances. On the other hand, the application of external sensors like ultrasonic modules raises the problem of hardware being more complicated and might also lead to certain questions being raised as to the system's portability and maintenance. The architecture based on CNN, despite being very good in terms of accuracy, still demands a considerable amount of power which makes its usage on low-power mobile devices impossible. The focus of the study is on short-range obstacle detection which is not accompa-

nied by the high-level understanding of pathway estimation or semantic scene interpretation. Moreover, the feedback system is limited to one type of input only, that is audio, which may pose problems in noisy environments or for users with hearing loss.

3.2.14.3 Relationship to the Proposed Research Work

The methodology that is presented by Masud et al. [**?**] is very much in line with the intentions of the suggested project because it deals with the detection of obstacles in real time and gives the user feedback for navigation of the visually impaired. Their CNN-based recognition underscores the necessity of computer vision as the main feature of mobility systems for assisting movement. In contrast with Masud's system utilizing several sensors and hardware-specific parts, Sighteo depends solely on the smartphone camera, thus making the solution more affordable, widespread, and simple in deployment. Moreover, the use of haptic feedback in the proposed system also complies with the use of auditory feedback utilized in Masud's work, thus enhancing the usability in varied environments. The current research findings apart from verifying the significance of the fusion among multiple sensors and highly precise object identification also indicate the requirement for light but reliable smartphone-based systems without compromising hardware complexity.

3.2.15 Low-Cost Multifunctional Assistive Device for Visually Impaired Individuals

This work describes the design of a low-cost wearable that can deliver a range of assistive functions integrated on one platform: text reading, obstacle warning, and remote assistance. Thus, it is a multifunctional tool that can serve various daily needs. The authors significantly stress that such systems must be affordable and practical. They emphasize that design decisions are done in support of real-world deployments rather than experimental work alone.

3.2.15.1 Summary of the Research Item

Kral et al. [**?**] present a pair of smart glasses. Their technology merges text recognition, obstacle detection, and distant help into one economical packing. The setup is organized around a Raspberry Pi Zero 2W, with an Arducam IMX519 autofocus camera, and operated through tactile button interfaces for users' convenience. On the spot audio feedback notifies users of the presence of objects, and recognized text is read to the user making the user alert in the situation. The gadget is wanted at a rehabilitation center, where the testing results showed very high accuracy in text recognition, and controlled settings gave better navigation support. The authors have also compared their approach with that of commercial products and have claimed that their solution is more attractive and less bulky. They point out limitations in processing speed, hardware size, and the future of their research includes investigating haptic feedback

and AI-driven scene analysis.

3.2.15.2 Critical Analysis of the Research Item

A key strength of this study is the integration of low-cost components like the Pi Zero and Arducam camera to provide multiple assistive functions like text recognition, obstacle detection, and remote support into a single wearable device. Its testing was done at a rehabilitation center, which adds to its credibility. However, its reliance on audio feedback alone limits usability in noisy environments, and the bulky, power-hungry hardware may affect comfort and battery life. The system also faces latency issues due to limited processing power. The authors note that future work should focus on improving speed, miniaturization, and broader field testing with visually impaired users.

3.2.15.3 Relationship to the Proposed Research Work

This article is very much in line with the objectives of Sighteo: both intend to provide real-time navigation help via the use of wearable or mobile systems. The integrated design creates a picture of how Sighteo could merge the detection and feedback modalities into a small form factor. Besides that, Kral et al. gave us a hint that the price tag, choice of components, and system integration are the factors that will determine whether a product will be adopted or not. Their marking of the hardware and latency limitations has also given us a warning, which is quite valuable in the case of Sighteo, where we should focus on low-latency pipelines and may consider, as they do, the use of haptic feedback to support audio cues. Most importantly, their study does not deal with either the YOLO models or the inference at smartphone level, and thus Sighteo's application of YOLOv8 on a mobile platform can be seen as a more lightweight and flexible variant of their concept. To summarize, this paper contributes to the literature we have by revealing a real prototype implementation that we can learn from and also differentiate against.

3.3 Literature Review Summary Table

Table 3.1: Summary of Literature Review

Author	Methodology	Results	Limitations
See et al. [?]	Smartphone depth imaging; 23 keypoints; zone-based obstacle mapping	>80% obstacle identification; effective multimodal alerts	1.6 m range; static evaluation; no path planning
Salunkhe et al. [?]	TensorFlow Lite SSD MobileNet on Android; TTS feedback	∼90% on-device detection accuracy; real-time audio cues	No continuous guidance; degrades in clutter/low-light
Ghahremanians et al. [?]	Low-latency pathfinding algorithm for safe routing	Fast route recomputation; improved traversal time	Robustness unproven indoors
Zhang et al. [?]	YOLOv5 + GhostNet + BiFPN with depth camera; audio output	42.4% model-size reduction; +1.2% recall vs. baseline	Requires depth sensor; controlled setting
Abidi et al. [?]	Review of CNN/LiDAR/ultrasonic and mobile aids	Identified method categories; mapped sensor/feedback trends	Descriptive synthesis; no experimental validation
Okolo et al. [?]	Survey of AI/ML/IoT assistive navigation pipelines	Reported integration patterns across sensing/feedback	Theoretical focus; lacks quantitative metrics
Casanova et al. [?]	PRISMA-based systematic review (58 studies)	Consolidated strengths/weaknesses; adoption insights	Affordability and usability gaps persist
Patel et al. [?]	Review of ML/DL for accessibility	Taxonomy of techniques and use-cases	No device-level deployment analysis
Al Duhayyim [?]	EDLES-SIAM: ABF + ResNet50 features + BiLSTM/SSAE/DNN ensemble with IoT	99.25% (UR Fall Detection) for HAR/fall detection	High compute; IoT infra required; no user study
Chavan et al. [?]	Quantized Florence-2 (4-bit) on Jetson Orin Nano; distance-aware VLM; Parler TTS Mini	∼5-frame latency; contextual, distance-aware audio	Edge-device dependency; limited-scale eval
Pfitzer et al. [?]	MR.NAVI on HoloLens2; integrates EOA/ETA/-PLD; spatial audio + NLP	Improved user confidence/navigation awareness	Expensive hardware; lighting sensitivity
Breve et al. [?]	CNN + transfer learning + Particle Competition/Cooperation	Higher classification accuracy than plain CNNs	Computationally heavy; no real-time user trials
Thandu et al. [?]	NLP chatbot over VEO-optimized indoor nodes for routing	High response accuracy; reduced disorientation vs. static prompts	Needs pre-mapped nodes; scalability limits
Masud et al. [?]	CNN vision + ultrasonic fusion; audio alerting	Accurate short-range detection; low-latency trials	Added hardware complexity; audio-only feedback
Kral et al. [?]	Pi Zero + Arducam wearable; DL for OCR/object detection; remote support	Integrated text/object features; functional prototype	Bulky; latency; audio-only interface

3.4 Conclusion

The literature review highlights that while AI-based navigation technologies have progressed significantly, each approach presents trade-offs. Research studies showcase strong accuracy and innovative techniques, but often rely on expensive hardware or lack real-world testing. Smartphone-based works improve accessibility but are mostly restricted to object recognition without full navigation support. In conclusion, despite the extensive research on the topic, a portable, affordable, and a truly navigational system has not been developed as of now. Our proposed system aims to directly address these challenge by relying solely on smartphone cameras and providing real-time feedback through audio and haptic cues.

Chapter 4 Software Requirement Specifications

A descriptive account of software requirements for the proposed AI-based navigation system, Sighteo, has been presented in this chapter. The functional and non-functional requirements, key features, and quality attributes of the system are outlined in chapter. It documents how the system will operate, the services it will provide, and the constraints under which it must function. Furthermore, this chapter makes sure that the requirements are precise, testable, and serve as a solid base-line foundation for the system design and future development. The details of specifications documented in this chapter provide a comprehensive understanding of what the software must achieve to effectively address the needs of its users.

4.1 List of Features

Our mobile app aims to revolutionize navigation for visually challenged users by providing the following features:

- Smartphone camera can be used by the users to receive real-time obstacle detection.
- Users can receive auditory feedback for safe navigation.
- Users can receive haptic (vibration) feedback as an alternative guidance mechanism.
- Users can access indoor navigation support through camera-based detection.
- Users can maintain a personalized profile for preferred feedback mode (audio/haptic).
- The system can adapt to different lighting and environmental conditions.
- Administrators can update and maintain training datasets for improved detection.
- The system can notify users of obstacles dynamically as they move.

4.2 Functional Requirements

Along with basic signup and login functionalities, the proposed application provides the following features to support visually impaired users:

- The system shall capture live video feed using the mobile device camera.
- The system shall process the captured video on device to detect obstacles, paths, and surrounding objects.
- The system shall provide real-time feedback (audio or haptic) to guide the user safely.

- The system will alert for unforeseen or moving objects (e.g. cars, humans, animals).
- The system will enable the users to report issues that they have confronted (e.g. incorrect identification of objects or inappropriate navigational help) simply without any complex fields.

4.3 Quality Attributes

The purpose of the application is to provide a secure, reliable, and easy-to-use navigation solution. It meets the needs of its users in such a way that the following quality attributes emerge prominently, providing the basis for defining the non-functional requirements:

- **Reliability:** Assures correct obstacle detection and reliable feedback.
- **Security:** Safeguards user data and guarantees secure communication within the application.
- **Usability:** Designed with simple interaction flows suitable for visually impaired users.
- **Correctness:** Object detection and navigation guidance with no errors is provided.
- **Flexibility:** Audio and haptic feedback based on user preference is provided.
- **Scalability:** System would be extensible to support more users in the future.

4.4 Non-Functional Requirements

- The system shall navigate in a reliable manner without unexpected failures to provide navigation at all times.
- The system shall provide data integrity, preventing data loss or corruption during in-application communication.
- The system shall have an intuitive and user-friendly interface, with should be easy for visually impaired individuals to navigate.
- The system shall provide exact and real-time navigation guidance and detection of objects.
- The system shall validate system-generated feedback to ensure that outputs (voice, sound, or haptic) are correct and consistent.
- The system shall be scalable to handle a growing number of users, data streams, and real-time requests without compromising performance.

4.5 Assumptions

- Only indoor enviroments.
- Proper lighting conditions.
- The smartphone is carried correctly at a proper angle.
- Walking at a normal speed.
- Adequate smartphone hardware.
- Train before using the app in public settings.

4.6 Use Cases

Table **??** onwards provide use cases for the project.

Table 4.1: Register User Use Case

Name	Register User		
Actors	User		
Summary	The user shall provide their information on the registeration form, and after successful verification, redirect the user to the login page.		
Pre-Conditions	The user must not be in the database records.		
Post-Conditions	The user is successfully registered and shall be redirected to the login page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the registration page.	2	The registration page is displayed asking for user's information.
2	The user enters all the email and information.	4	The system verifies the information, and redirects the user to the login page.
Alternative Flow			
3	The user enters existing email.	4-A	The system responds with an error message: Account already exists

Table 4.2: Login User Use Case

Name	Login User		
Actors	User		
Summary	The user shall provide their email and password on the login form, and after successful verification, redirect the user to the home page.		
Pre-Conditions	The user must be in the database records, either added by any of the authorized users or added manually by a developer. The user must not already be logged in.		
Post-Conditions	The user's session is successfully established and shall be redirected to the home page.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user opens the login page.	2	The login page is displayed asking for email and password.
2	The user enters valid email and password.	4	The system verifies the email and password, establishes a session for the user and redirects the user to the home page.
Alternative Flow			
3	The user enters invalid email or password.	4-A	The system responds with an error message: Incorrect email or password entered.

Table 4.3: Navigate User Use Case

Name	Navigate User		
Actors	User		
Summary	Enables a visually impaired user to receive real-time guidance for safely moving through indoor environments using the app.		
Pre-Conditions	The user is logged into the app with a functional camera and sensors, has granted required permissions, and is in an environment with adequate lighting.		
Post-Conditions	The app successfully guides the user, ensuring safe navigation and providing feedback if obstacles or issues are detected.		
Special Requirements	None		
Basic Flow			
Actor Action		**System Response**	
1	The user clicks the navigate button.	2	The live camera feed starts to show on the screen.
2	The user faces an obstacle.	4	The system successfully identifies the obstacle and provides guidance with a way around it.
Alternative Flow			
3	The user presses the back button.	4-A	The system goes to the home page.

4.7 Hardware and Software Requirements

This chapter presents a descriptive account of the software requirements for the proposed AI-based navigation system, Sighteo. The chapter discusses the functional and non-functional requirements, key features, and quality attributes of the system. It documents how the system will operate, the services it will provide, and the constraints under which it must function. Furthermore, this chapter ensures that the requirements are precise, testable, and provide a sound base-line foundation for the system design and development in the future. The details of specifications documented in this chapter give full insight into what the software must accomplish to meet its users' needs effectively.

4.7.1 Hardware Requirements

The hardware requirements have been kept minimal to ensure accessibility of the application on standard mobile devices. The system is designed to run on smartphones equipped with a functional rear camera, speakers, and vibration motor to support voice, sound, and haptic feedback. Moderate processing power and memory are sufficient since the heavy computations will be simplified to run on standard smartphones. However, the device must maintain a stable internet connection to enable uninterrupted video streaming and real-time feedback. That is one of the key requirements. Due to the responsive nature of our application, it will be able to adapt to different screen sizes and resolutions. This responsiveness will also make sure that Sighteo is made accessible to users having a wide variety of smartphones or tablet devices with varying screen sizes.

4.7.2 Software Requirements

It will be recommended to operate our application, Sighteo, on a smartphone with an updated mobile operating system (OS). This is a requirement for making the experience seamless, free from system level lags, and to ensure that all features behave as intended. The device must support background services for camera access, audio playback, and vibration for the application to function properly, and give navigation cues. Equally, keeping the application updated is encouraged. Like any other application, keeping Sighteo updated would ensure that it includes compatibility, security patches, and possible performance enhancements. Activating the in-built accessibility settings like screen readers or enhanced audio settings can further improve the experience. This consideration would mean that Sighteo could play a truly practical role in everyday life. All these software requirements intend to make the user's navigation experience smooth, consistent, and reliable when using Sighteo.

4.8 Graphical User Interface

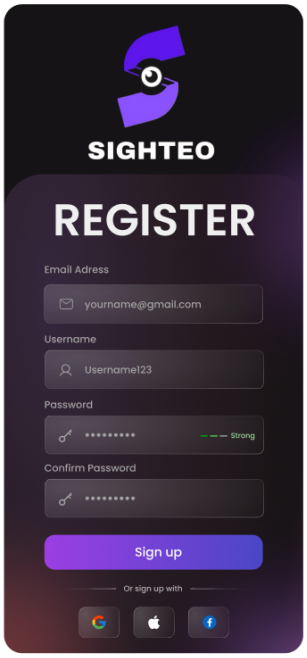

Figure 4.1: User Registration Page

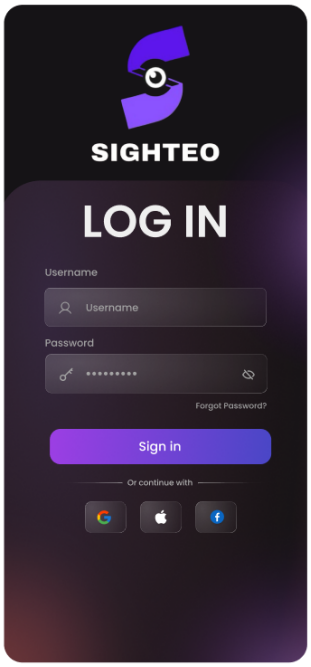

Figure 4.2: User Login Page

Figure 4.3: Navigation Page

4.9 Database Design

Our database schema is modular in design. The user profile information is stored in the user table, while the contact information has been divided for ease into the contact details table. The login credentials are securely kept by the authentication table for proper account management. Lastly, the user feedback and ratings related to the app are kept in the feedback table. In designing this schema, our intention was to keep it simple, avoid redundancy so each tableshould play a specific, dedicated role.

4.9.1 ER Diagram

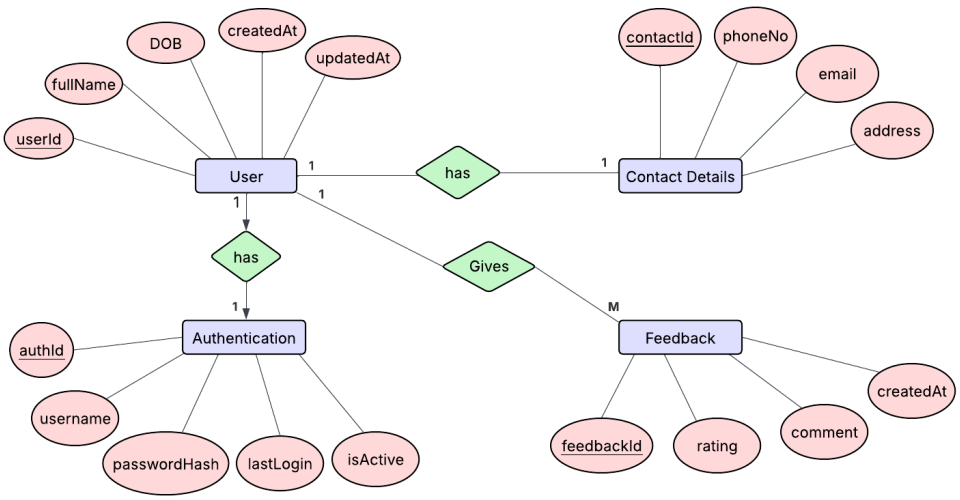

Figure 4.4: Entity Relationship Diagram

4.9.2 Data Dictionary

Table 4.4: Data Dictionary

Entity	Attribute	Data Type	Relation To	Nullable	Description
User	”userId”	int		No	Unique ID (PK)
	fullName	string		No	Name of User
	DOB	Date		No	Date of Birth
	createdAt	Date		No	Date of User creation
	updatedAt	Date		No	Date of last Update
ContactDetails	”contactId”	int		No	Unique ID (PK)
	userId	int	User	No	corresponding User
	phoneNo	string		No	phone number of User
	email	string		No	email of User
	address	string		Yes	address of User
Authentication	”authId”	int		No	Unique Id (PK)
	userId	int	User	No	corresponding User
	username	string		No	username of User
	passwordHash	string		No	Hashed password of User
	lastLogin	string		No	When User last Logged in
	isActive	boolean		No	Is the User active
Feedback	feedbackId	int		No	Unique ID (PK)
	userId	int	User	No	corresponding User
	rating	int		No	Rating given by User
	comment	string		Yes	Comment about the experience
	createdAt	Date		No	When was Feedback created

4.10 Risk Analysis

Identification of the potential risks and uncertainties likely to impact project progress is a very crucial activity in the process of project planning. In order to ensure that the development process runs smoothly and the project is delivered successfully, risk analysis of our system considers many aspects with the purpose of efficiently counteracting possible problems.

- **Hardware Failures:** Although the application has low hardware requirements, some unexpected device failures (like camera failure, malfunctioning speakers, or malfunctioning vibration motor) may impede service availability. Routine maintenance of the devices and fallback notifications are recommended to counteract this risk so that users are always in time notified in case essential hardware is not available.

- **Software Compatibility:** The application requires recent versions of mobile operating systems (Android/iOS). Camera usage, audio playing, or background services may be incompatible due to unsupported/older versions. Regular updates of apps and operating systems forced on users can decrease this risk.

- **User Experience:** The target users in this case are visually impaired people, therefore, accessibility is, the most crucial aspect. Usability can considerably reduce due to complicated or illogical interfaces. A simple audio-sound-based design with vibrotactile feedback will eliminate the risk by ensuring users receive clear and consistent instruction without any kind of visual interaction.

4.11 Conclusion

The aim of this chapter was to formalize the functional and non-functional requirements that will set a basis for developing Sighteo. System features, performance expectations, hardware and software needs, and quality attributes make the application reliable, usable, and safe for visually impaired users. This chapter translates the users' needs into structured requirements and hence provides an exact blueprint for subsequent design and implementation phases in such a way that each component of the system is aligned with the core purpose of the project.

Chapter 5 Proposed Approach and Methodology

This chapter sets out the general approach informing the design of our assistive navigation system, Sighteo. It describes how the application is structured, how the core models are chosen and integrated, and how user feedback informed every design decision. The following sections outline the methodology, tools, and ethical considerations guiding the project, along with the long-term directions beyond the initial indoor navigation focus.

5.1 Overview of the Proposed Approach

Creation of a mobile application that aids visually impaired people in the navigation of their daily environment with the help of a smartphone's camera alone, rather than earlier solutions that were reliant on external sensors like LiDAR or GPS, is our primary goal. Our system focuses on accessibility and affordability by making use of computer vision (CV) models running on the backend. Real-time video frames, captured by the smartphone camera, are sent to the backend for processing. The backend processes the input frame by making use of object detection and vision-language models, then sends audio and haptic feedback to the user.

5.2 Methodology

Iterative model development combined with software engineering best practices is the methodology adopted for this project's development. Our approach comprises of following phases:

1. **Requirement Analysis and Data Collection:** Datasets comprising of annotated objects (doors, stairs, obstacles, furniture) present in an indoor environment must be collected. Interviewing with visually impaired users and accessibility experts can help in defining user-centric requirements.
2. **Preprocessing and Model Selection:** Input data for object detection must be preprocessed alongwith semantic segmentation tasks. A suitable pre-trained vision-language model such as YOLO-based frameworks with depth estimation must be selected.
3. **System Architecture Design:** Mobile frontend developed using React Native, ensuring compatibility across both iOS and Android. Backend implemented in Python for AI inference models.
4. **Integration of Feedback Mechanisms:** Audio feedback by making use of text-to-speech modules to announce detected objects and provide verbal guidance. For urgent alerts like approaching obstacles, non-verbal cues will be given like beeps and vibration patterns.
5. **Testing and Evaluation:** Volunteer test subjects can be employed for testing the system in a

controlled indoor environment. Detection rate, response time, and user satisfaction ratings can be the test metrics. Based on test results, iterative optimization can be performed.

5.3 Agile Development

Agile Development Approach to allow for incremental improvements and feedback from users will be followed. Our main markers are initial prototype, integration of object detection, testing of feedback mechanism, and usability tests, upon which sprints will be based. Flexibility and user-initiated changes are taken care of by Agile.

5.4 Tools and Technologies

- **Frontend:** React Native with TypeScript
- **Backend:** Python
- **AI Models:** Pre-trained object detection models (e.g YOLO) with fine-tuning to support important obstacles like walls, stairs etc.
- **Database:** PostgreSQL (or any other database service).

5.5 Ethical and Accessibility Considerations

Particular attention will also be given to accessibility since the target members will be visually impaired individuals. Adherence to the level of provision of the WCAG standards, protection of the privacy of the taken photographs, and omission of unnecessary user inputs belong to this. In order to ensure ethical considerations and inclusivity, the system will be tested with the involvement of real users.

5.6 Future Directions

Although the initial focus is on indoor navigation in controlled environments, the methodology helps us to progressively extend the scope of the project in the future. The project roadmap envisions:

- Semi-controlled indoor environments (e.g., public buildings with moderate variations).
- Dynamic indoor environments with moving objects and unpredictable layouts.
- Outdoor navigation as the final extension, incorporating distance-aware detection in open, unconstrained settings.

5.7 Conclusion

In short, the methodology blends AI object detection and real-time feedback into a mobile application. It begins with the selection of a suitable base model from out of YOLOv8, SSD and RetinaNet. After selection, we'll move to fine tuning the model to the specific needs of the vusially impaired individuals. Scalability, affordability, and accessibility are offset in the project by the utilization of lightweight on-device computation managed with help of Python and React Native. Agile development model allows for iterative fine-tuning. Aim is to increase inclusivity and trust in the visually challenged community. This will be ensured through ethical guidelines.

Chapter 6 High-Level and Low-Level Design

The following chapter describes the high-level and low-level design for Sighteo and explains how different components work together to support indoor navigation in real-time. This includes a high-level description of the major architectural choices, system modules, and the flow of data from the camera down to the feedback layer. The provided details form the technical backbone of the application. In this way, the chapter sets up the structure that will be needed during implementation in later phases.

6.1 System Overview

The system allows visually impaired people to navigate indoor environments safely and independently. The procedure includes system architecture designing, preparing and fine-tuning datasets, training the detection model, integrating the trained model into the mobile application, and providing audio and haptic feedback to the user.

6.1.1 System Design

Sighteo's system is organised into a set of interconnected modules that work together to deliver the final objectives of the application. Each module handles a specific part of the workflow. Following describes the compositional structure of the system:

6.1.1.1 User Authentication

The authentication module is the point where users start their experience. It contains signup and login functionality to ensure secure access. This module validates users' credentials, and smoothly transitions them to the navigation interface.

6.1.1.2 Navigation Control

The navigation module is the core functionality of the application. It allows users to control when to start and when to stop navigation from the homepage. When navigation is activated, the trained AI model is the fed real time video frames from the camera, which provides navigational cues as the output.

6.1.1.3 Object Detection and Feedback

The object detection module processes the video frames from the camera using the AI model, which would be selected after ample experimentation and considering trade-off between accuracy and speed. The model identifies obstacles in real time, and the user is alerted through audio and haptic feedback.

6.1.1.4 Feedback

This module provides users with an option report an any issue that they encounter using the application. A button will be provided on the homepage for it. This will provide endless opportunities for application refinement.

6.1.2 Data Collection

For detection model training, datasets containing everyday objects present in a common indoor setting will be collected. Techniques for data augmentation like rotation, scaling, and brightness will be used for enhancing diversity and strength. These procedures ensure that the model will perform efficiently under varying environmental conditions such as varying light or object orientation.

6.1.3 Model Training

The training phase resolves around preparing the dataset collected for the detection model. The dataset is annotated with object classes for indoor navigation. The AI model will be trained and fine-tuned to provide a balance between detection and inference speed. Experimental comparisons would be conducted among different AI models and their respective variants to make sure that the chosen model is the most stable and reliable. The chosen model is then trained and integrated into the application to provide the best navigational output as possible.

6.2 Design Considerations

The following are the key concerns that should be identified prior to the finalization of the Sighteo system design: These help in pointing out various technical limitations, user experience expectations, and environmental issues that could control how an app would function in real-world situations. By analyzing such concerns at an early stage, the design should remain practical and scalable, and meet the needs of visually impaired users by ensuring that each and every design decision supports the requirements for safety, reliability, and consistency in navigation.

6.2.1 Assumptions and Dependencies

The design of Sighteo relies on the following assumptions and dependencies to ensure proper functioning:

- **Related software or hardware:** Smartphones would be not have outdate software, have functional cameras, speakers and vibration motors for feedback.

- **Operating systems:** The application would be compatible with Android and iOS.
- **End-user characteristics:** The user would have the fundamental knowledge of how to use and operate a smartphone and the applications installed on it.
- **Possible and/or probable changes in functionality:** The application considers the possibility of introducing newer object detection models in the future, or different ways of feedback to the user.

6.2.2 General Constraints

Sighteo's design is shaped by various constraints, each significantly impacts the system's software architecture and functionality.

- **Hardware or software environment:** Sighteo depends on mobile devices equipped with functional cameras and vibration hardware. Low-end devices may experience slower response times due to limited processing power or outdated hardware.
- **End-user environment:** Object detection can be hindered due to usage in environments with poor lighting, cluttered spaces, or fast moving objects.
- **Availability or volatility of resources:** The device's resources such as battery life, camera quality, and processing speed play a major part in system's performance.
- **Standards compliance:** The design must show compliance with accessibility standards for visually impaired individuals. It should also obey platform specific guidelines like Google play and Apple app store requirements.
- **Interoperability requirements:**The application must work smoothly with the trained YOLOv8 model and mobile device hardware. Interoperability in the future could be expanded to cloud-based updates or third-party accessibility services.
- **Interface/protocol requirements:** The user interface must be simple ensuring easy transition between the login, main page, and feedback page. Communication protocols between app modules (camera feed, feedback delivery, etc) should work efficiently in real time.
- **Data repository and distribution requirements:** User account information must be safely stored in a password-protected database. Data distribution, such as model updates, must go via secure channels to prevent security issues.
- **Security requirements (or other such regulations):** The system must protect sensitive information, including login credentials, from unauthorized access. Data in storage and transmission should be encrypted to avoid tampering.

- **Memory and other capacity limitations:** Mobile devices with limited RAM or processing capacity may face difficulty in running the app efficiently and smoothly. Therefore, the size of the model must be optimized to reduce overloading the device.
- **Performance requirements:** The system must deliver real-time feedback. It should be able to target at least 15–20 frames per second for obstacle detection to ensure safe navigation with audio and haptic feedback.
- **Network communications:** Even though navigation process does not require internet, internet connectivity may be required for updates and feedback posting. These services could be delayed due to disruption in internet.
- **Verification and validation requirements (testing):** Testing will be performed in controlled environments to ensure detection accuracy, timely feedback, and system dependability prior to widespread deployment.
- **Other means of addressing quality goals:** Frequent user testing with visually impaired users will improve usability. Ongoing updates and patches will remedy bugs and enhance accuracy.
- **Other requirements described in the requirements specification:** Additional requirements may include multi-language support in the help system and scalability for integration with future external sensors or wearable devices.

6.2.3 Goals and Guidelines

Sighteo is designed around a number of goals, directives, principles, and priorities for effectiveness, reliability, and user-friendliness of its design for visually impaired users.

- **The KISS principle ("Keep it simple stupid!"):** Sighteo adoption requires simplicity. Only essential functions like Start/Stop Navigation and Help are placed front and center on the home page. This minimizes complexity and allows users to use the app easily and confidently.
- **Emphasis on speed versus memory use:** Sighteo is focused on low-latency performance. After experimentations and testing, YOLOv8 was selected due to the best blend of accuracy and speed.
- **Accessibility and user-centered design:** It is very important that the application must be as simple as possible with haptic feedback for accessibility. This will avoid users being confused with the interface.
- **Reliability and safety:** As a safety assistive tool, safety is of paramount importance. In order to make the users trust the application, design guidelines have been followed strictly to allow consistent and reliable detection and feedback.

- **Scalability and adaptability:** Currently, Sighteo is aimed for indoor navigation, but the development will done in such a way to integrate outdoor navigation in the future.

6.2.4 Development Methods

The development of Sighteo follows a systematic and well-defined approach to ensure reliability, accessibility, and adaptability.

- **Agile Development Methodology:** Agile approach has been used in Sighteo to handle changing requirements and make the development stage more flexible. Work is divided into sprints, concentrating on modules like authentication, navigation control, and feedback mechanisms. The process ensures continuous improvement via incremental posting.
- **Iterative Design Process:** The system uses an iterative design cycle to improve core functionalities. Testing and feedback are taken into account in each iteration so that the navigation experience becomes more and more accurate, responsive, and user-friendly.
- **User-Centered Design (UCD):** Sighteo is built for visually impaired individuals, therefore, UCD principles are vital to its development. The highest priority has been given to feedback mechanisms, interface simplicity, and accessibility features such as haptic alerts as this ensures that the system aligns with the expectations and safety requirements of its target audience.

6.3 System Architecture

The architecture of the system is structured into three interconnected layers: presentation, logic, and data. The presentation layer consists of a mobile application that is the main point of interaction between the user and the system. It uses the camera to capture real-time video frames and delivers processed results back to the user as audio or vibration feedback, making it easier for visually impaired individuals to respond to their surroundings. The logic layer is the processing hub of the system. At this level, the video frames are passed through the object detection model, which detects obstacles in the environment, and the navigation logic, which determines how the user can avoid these obstacles and move safely. Lastly, the data layer stores vital information such as user details, credentials, and feedback. This ensures secure access to the system while allowing continuous improvement based on user experiences. Together, these layers provide a smooth workflow for reliable and real-time navigation.

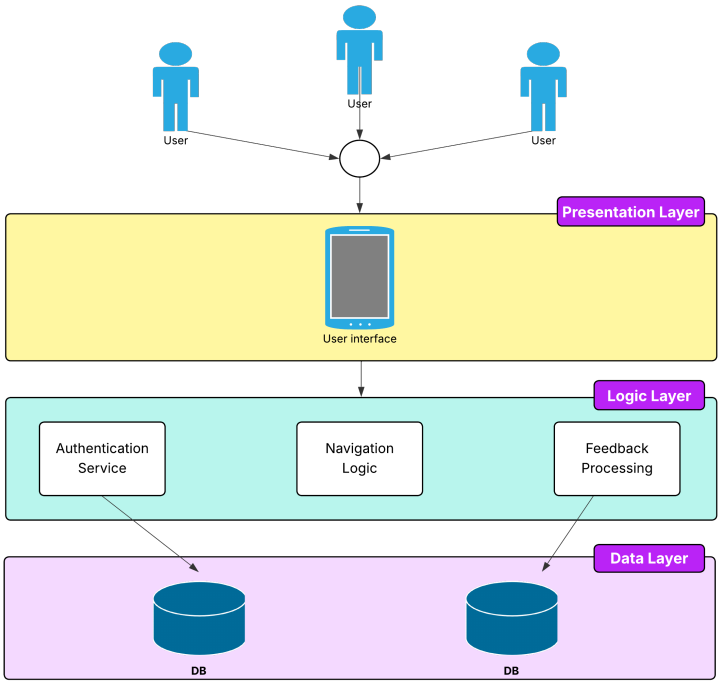

Figure 6.1: High Level System Architecture Diagram

6.3.1 Subsystem Architecture

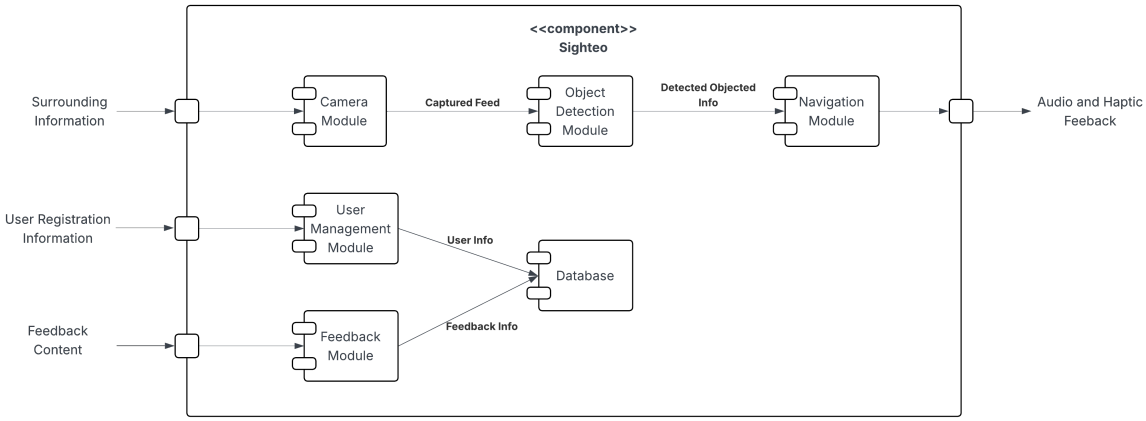

Figure 6.2: Sub-System Architecture Diagram

Sighteo's subsystem architecture is organized into interconnected modules that work alongside each other to provide navigation cues. The camera module captures real-time video, which is sent to object detection module as video frames for obstacle identification. This information flows into the navigation module, which calculates safe paths and returns instructions to the user through audio and haptic feedback. The User management module handles authentication and stores sensitive data in the database, while the feedback module stores the experiences shared by the users, which are then used for analysis. Together, these subsystems ensure reliable, responsive, and user-centered navigation support.

6.4 Architectural Strategies

The architectural strategies for the Sighteo app focus on providing dependable real-time assistance to visually impaired users while establishing cross-platform compatibility, system scalability, and strong security measures. The mobile app is developed in **React Native**, while all the backend will be built in python. This division makes sure that the application runs light and remains responsive on mobile devices, while the backend efficiently manages AI workloads and heavy computations.

6.4.1 Tools and Software

This section describes the technical basis on which Sighteo can function reliably in real time. It summarizes the major software components, hardware requirements, flow of communication, and safety features that are integrated into the overall design.

6.4.1.1 Mobile Application:

- Developed using **React Native**, enabling simultaneous compatibility with both iOS and Android.
- Integration with native accessibility APIs such as iOS VoiceOver and Android TalkBack.
- Usage of device features like Text-to-Speech (TTS), vibration motor for feedback.

6.4.1.2 Backend Services:

- Implemented in **Python** to gain access to newer and better libraries and packages
- Machine Learning inference powered by **TensorFlow** or **PyTorch**.
- Middleware layer for request handling and load balancing.

6.4.1.3 Communication Protocols:

- Lightweight payloads (JSON/protobuf) carry detection outputs such as object type, relative distance, and confidence scores.

6.4.2 Hardware

Client Devices: Smartphones with a working rear camera and vibration motor.

6.4.3 Communication Mechanisms

- The mobile client captures video frames at a preset rate based and sends them securely to the backend.
- The backend processes the frames, performs inference, and sends back results for instant playback.

6.4.4 Error Detection and Recovery

- Backend implements retry mechanisms in case of internal errors.
- Uncertainty within model detection is communicated conservatively (e.g., “possible obstacle ahead”) instead of remaining silent.

6.4.5 Security and Data Privacy

- All API calls will be secured by secrets.
- No video frames are permanently stored on the backend as results are delivered in real-time.

6.4.6 Future Plans

- Integrate traffic signal recognition, obstacle trajectory prediction, etc in the future.
- Enhance multi-language voice support to better serve diverse user groups.

6.5 Domain Model/Class Diagram

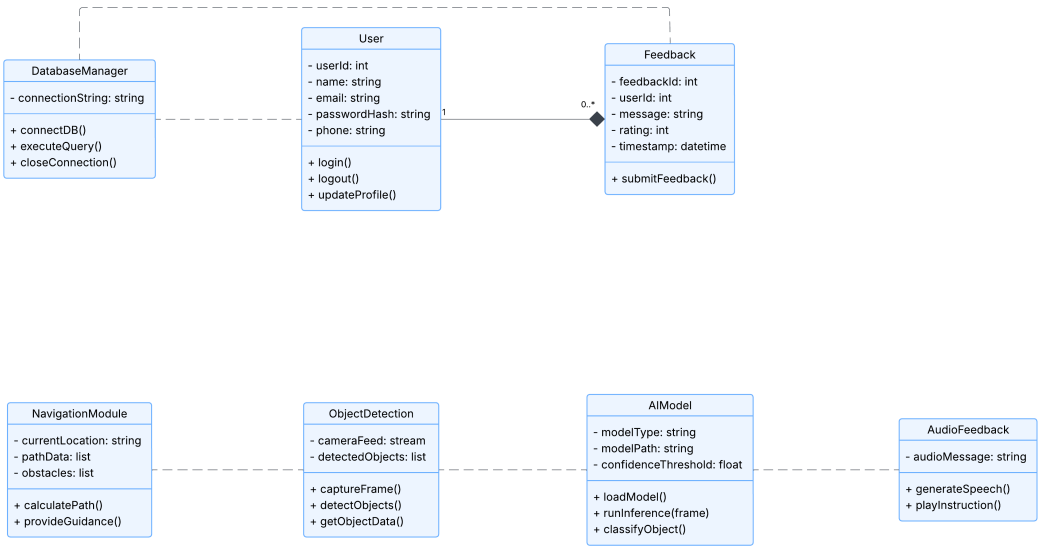

Figure 6.3: Class Diagram (Tentative)

6.6 Policies and Tactics

This section describes the design policies and strategies used during Sighteo development. These choices, though not affecting the system's top-level architecture significantly, affect implementation details, coding style, testing, and maintainability in the long term.

6.6.1 Conventions and Coding Guidelines

For Sighteo development, coding standards shall be strictly adhered to ensure readability, consistency, and maintainability. The mobile app will be built using React Native, whereas backend and integration of models will be done using Python. Proper indentation, module-based coding standards, and explanatory comments shall be used throughout the codebase. This makes it easy for future developers to comprehend and extend the project.

6.6.2 Coding Environment

The primary Integrated Development Environment (IDE) will be Visual Studio Code due to its extensive support for React Native and Python. It has built-in debugging tools, syntax hints, and extensions for version control like github. Furthermore, to manage Python environments during model training and experimentation, Anaconda will be used.

6.6.3 System Testing

White-box and black-box test methodologies will both be employed in order to ensure the reliability of the system. Black-box testing will be used to test user interactions such as signin/signup, control navigation, and reaction to feedback, while white-box testing will ensure that modules like frame processing and YOLOv8 inference are correct. Unit tests and integration tests will be executed before deployment.

6.6.4 Maintenance of Software

Once the project is complete, bug fixes, dependency upgrades, and added features like support for semi-controlled outdoor spaces will be part of maintenance. Modular design will make it possible to update without halting the whole system.

6.6.5 Extensibility

Sighteo is extensible. Its code is written in a modular way such that future developers can easily plug in new object detection models, other sensors, or different feedback mechanisms. This makes it possible for the system to continue evolving with technological progress and user requirements.

6.6.6 Engineering Trade-offs

A major design trade-off in Sighteo is the selection of YOLOv8 over more recent versions such as YOLOv10. Although more recent versions could potentially provide slight improvements, YOLOv8 has been chosen because it has an established balance of being fast, accurate, and stable, and hence is more reliable for real-time navigation applications on handheld devices.

6.6.7 Requirements Traceability

To ensure that all requirements are addressed, a mapping between functional requirements and implemented modules will be maintained. This traceability will help verify that user needs and system goals are fully met during development and testing.

6.6.8 Deliverables and Build Process

The system deliverables will include the mobile application for both android and iOS, and documentation of the trained YOLOv8 model. The build process will involve compiling React Native code, linking it with the backend services.

6.7 Conclusion

The architecture, subsystem interactions, design principles, and technical strategies that constitute the Sighteo system were introduced. Particular details on how the application processes data, manages user interactions, integrates the detection model, and presents real-time feedback have been provided. This chapter has established that the system is based on a scalable, maintainable, and user-centered foundation by identifying assumptions, constraints, and the ways in which it has been developed. It is these types of design decisions that provide the backbone for the implementation work continuing in successive chapters.

Chapter 7 Implementation and Test Cases

Work done in terms of implementation of the application has been described in this chapter, including technical details which are not discussed in the high and low-level design.

7.1 Implementation

The implementation phase contained two main components. Firstly, a methodical model selection and fine-tuning process was undertaken. Secondly, the finalized model was integrated in an intuitive and functional React Native mobile application for real time navigation assistance.

7.1.1 Model Selection and Fine-Tuning

This phase was devoted to the identification and optimization of an object detection architecture that can achieve real time indoor obstacle detection on mobile hardware.

7.1.1.1 Model Exploration

Three object detection architectures (YOLOv8, SSD, and RetinaNet) of the present time were evaluated. Their selection was based on their documented performance in real time detection tasks. Each of the architectures was examined for their detection accuracy, inference speed, computational complexity and deployment feasibility in mobile devices. Particular focus was placed on achieving real time frame rates suitable for walking scenarios, as the latency directly affects user safety. Preliminary benchmarking showed that YOLOv8 had the most balanced trade-off between detection accuracy and inference speed.

7.1.1.2 Dataset Selection and Preparation

Publicly available datasets, namely COCO and Open Images, were used as the foundation for initial experimentation. Given that these datasets cover a wide range of object categories, a filtering process was added to keep the indoor relevant classes, such as chairs, tables, doors, stairs and people. Data augmentation techniques (resizing, rotation, brightness and random cropping) were used to mimic the indoor variability of the real world (e.g. lighting fluctuations, cluttered environments).

7.1.1.3 Evaluation Metrics

In order to compare models in an objective way the following metrics were used:

- **Mean Average Precision (mAP):** Mmeasures the quality of detection for the different classes of objects.

- **Frames Per Second (FPS):** Measures the performance capability in real time.
- **Model Size and Memory Consumption:** Measures the feasibility for mobile deployment
- **Latency and Power Consumption:** Measures responsiveness and resource efficiency

The general goal was to identify a model that maintained high detection accuracy and ensured real-time usability and efficient use of resources.

7.1.1.4 Platform and Experimental Setup

All the experiments were performed using Python based deep learning frameworks. Model benchmarking and visualization was performed with general machine learning libraries. Training and evaluation were conducted using a workstation with an Nvidia RTX 3060 GPU, and therefore consistent testing conditions were achieved across all the architectures.

7.1.1.5 Final Model Choice and Fine-Tuning

Following the methodology of comparative analysis, YOLOv8 was selected as the final architecture, which is attributed to the stable frame rate, small model size, and strong performance in different indoor lighting conditions. Subsequently, the model has been fine-tuned on the filtered indoor data set to improve the detection performance for the classes relevant for navigation. Hyperparameters were adjusted in an iterative manner to optimize convergence, detection accuracy, and inference stability.

7.1.2 Mobile Application Integration

After optimizing the model, the trained model (YOLOv8) was integrated into a React Native mobile application designed for visually impaired people.

7.1.2.1 Model Deployment

The trained model was converted to a mobile compatible format to make it easier to perform on-device inference. This conversion removed the need for cloud processing and therefore reduced the latency and increased the reliability in environments with limited connectivity.

7.1.2.2 Real-Time Detection Pipeline

The mobile application captured the frames from the camera continuously and these frames were sent to the embedded object detection model. Detected objects were processed to determine relative positioning and possible collision risk. A lightweight and rule-based logic layer gave priority to the obstacles that were most immediately dangerous.

7.1.2.3 Feedback Translation Layer

Detection outputs were converted into actionable advice through a feedback module. Audio prompts provided directional information, while haptic vibration patterns indicated proximity and urgency. This multimodal method ensured accessibility without causing cognitive overload.

7.1.2.4 End-to-End Testing

Integrated system testing was carried out in controlled indoor settings, comprising of corridors and rooms with obstacles. Performance measured from detection reliability, response latency and clarity of user feedback. Observations from testing led to iterative refinements in both the detection model and the mobile interface.

7.2 Test Case Design and Description

The test cases designed for Sighteo were aimed at testing the functional operation of the mobile navigation system in controlled indoor environments. Because the project is aimed at helping visually impaired users with obstacle detection in real time and guidance, the test cases were built around the principal user flows and system responses of the application.

All tests were run on Android mobile devices with operative camera access, speaker output, vibratory support, and adequate processing capability to run the integrated object detection model. The tests were conducted in an indoor environment, such as corridors, classrooms, rooms, and hallways, under normal lighting conditions, unless otherwise specified. For the tests related to navigation, it was assumed that the application had been installed properly, the necessary permissions had been granted, and that the user had already activated the navigation module.

The test cases have common attributes, i.e. valid user interaction with the mobile interface, correct loading of the camera feed, correct initialization of the detection model, and successful activation of the output modalities such as audio and vibration. The running of any individual test case is usually independent of any others, except where there is a prerequisite such as logging in or permission approval. The following test cases were created in order to check the main features implemented in Sighteo.

Table 7.1: User Registration

Authentication Module			
Sighteo Mobile Application			
Test Case ID:	TC-01	**QA Test Engineer:**	Hasnain Khurram
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Register User
Revision History:	None		
Objective:	To verify that a new user can register successfully in the application using valid credentials.		
Product/Ver/Module:	Sighteo / 1.0 / Authentication Module		
Environment:	Android smartphone, internet connection, Firebase/backend connectivity enabled.		
Assumptions:	The user provides valid input and the backend service is reachable.		
Pre-Requisite:	Application is installed and opened on the device.		
Step No.	**Execution Description**	**Procedure Result**	
1	User enters name, email, and password, then submits registration form.	The account is created successfully and the user proceeds to the next screen or login screen.	
Comments: The registration workflow operated correctly according to the system requirements.			
Passed			

Table 7.2: User Login

Authentication Module			
Sighteo Mobile Application			
Test Case ID:	TC-02	**QA Test Engineer:**	Hasnain Khurram
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Login User
Revision History:	None		
Objective:	To verify that a registered user can log into the application using correct credentials.		
Product/Ver/Module:	Sighteo / 1.0 / Authentication Module		
Environment:	Android smartphone with internet connection and application installed.		
Assumptions:	The user account already exists in the system.		
Pre-Requisite:	User has already registered successfully.		
Step No.	**Execution Description**	**Procedure Result**	
1	User enters valid email and password and taps login.	The user is authenticated and redirected to the home or navigation screen.	
Comments: Login functionality behaved as expected without errors.			
Passed			

Table 7.3: Camera Permission and Navigation Initialization

Navigation Module			
Sighteo Mobile Application			
Test Case ID:	TC-03	**QA Test Engineer:**	Hasnain Khurram
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that the application requests camera permission and initializes the navigation module successfully.		
Product/Ver/Module:	Sighteo / 1.0 / Navigation Module		
Environment:	Android smartphone with functional rear camera.		
Assumptions:	The application has been installed properly and model files are available locally.		
Pre-Requisite:	User is logged into the application.		
Step No.	**Execution Description**	**Procedure Result**	
1	User opens the navigation feature and grants camera permission if prompted.	Live camera feed opens and the navigation interface becomes active.	
Comments: Camera access and navigation startup completed successfully.			
Passed			

Table 7.4: Real-Time Obstacle Detection

Object Detection Module			
Sighteo Mobile Application			
Test Case ID:	TC-04	**QA Test Engineer:**	Hasnain Khurram
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that the system detects common indoor obstacles in real time from the camera feed.		
Product/Ver/Module:	Sighteo / 1.0 / Object Detection Module		
Environment:	Indoor room or corridor with obstacles such as chair, table, door, stairs, or person visible in frame.		
Assumptions:	Lighting is sufficient for the camera to capture usable frames.		
Pre-Requisite:	Navigation module is already active.		
Step No.	**Execution Description**	**Procedure Result**	
1	User points the device camera toward one or more indoor obstacles.	The system detects the obstacle(s) and updates the screen/output state in real time.	
Comments: Obstacle detection operated correctly for common indoor objects used during testing.			
Passed			

Table 7.5: Audio Guidance Response

Feedback Module			
Sighteo Mobile Application			
Test Case ID:	TC-05	**QA Test Engineer:**	Abdullah Azeem
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that the application provides audio instructions when an obstacle is detected.		
Product/Ver/Module:	Sighteo / 1.0 / Audio Feedback Module		
Environment:	Android smartphone with speaker or earphones enabled.		
Assumptions:	Volume is audible and text-to-speech output is configured correctly.		
Pre-Requisite:	Obstacle detection is running and at least one obstacle is present.		
Step No.	**Execution Description**	**Procedure Result**	
1	User approaches an obstacle while the navigation mode is active.	The system produces an audio message indicating obstacle awareness or directional guidance.	
Comments: Audio feedback was triggered correctly and was understandable.			
Passed			

Table 7.6: Haptic Feedback Alert

Feedback Module			
Sighteo Mobile Application			
Test Case ID:	TC-06	**QA Test Engineer:**	Abdullah Azeem
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that vibration-based haptic feedback is generated when a nearby obstacle requires user attention.		
Product/Ver/Module:	Sighteo / 1.0 / Haptic Feedback Module		
Environment:	Android smartphone with vibration support enabled.		
Assumptions:	Phone vibration settings are enabled and accessible to the application.		
Pre-Requisite:	Navigation mode is active and a relevant obstacle is detected.		
Step No.	**Execution Description**	**Procedure Result**	
1	User moves toward an obstacle at close range while holding the device.	The device vibrates according to the warning logic defined by the application.	
Comments: Haptic warning was issued successfully and matched the expected behavior.			
Passed			

Table 7.7: Safe Direction Suggestion

Guidance Module			
Sighteo Mobile Application			
Test Case ID:	TC-07	**QA Test Engineer:**	Abdullah Azeem
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that the system suggests a safer direction when an obstacle blocks the user's current path.		
Product/Ver/Module:	Sighteo / 1.0 / Guidance Module		
Environment:	Indoor corridor or room with obstacle placed in the forward walking path.		
Assumptions:	The model detects the obstacle correctly and the guidance logic is active.		
Pre-Requisite:	User is in navigation mode with audio feedback enabled.		
Step No.	**Execution Description**	**Procedure Result**	
1	User points the camera toward a blocked path.	The application issues a directional response such as moving left, right, or stopping.	
Comments: Direction guidance responded correctly for the tested blocked-path scenario.			
Passed			

Table 7.8: Navigation Session Stop

Navigation Module			
Sighteo Mobile Application			
Test Case ID:	TC-08	**QA Test Engineer:**	Abdullah Azeem
Test Case Version:	1.0	**Reviewed By:**	Farooq Haider
Test Date:	11/03/2026	**Use Case Reference(s):**	Navigate User
Revision History:	None		
Objective:	To verify that the user can stop the active navigation session safely.		
Product/Ver/Module:	Sighteo / 1.0 / Navigation Module		
Environment:	Android smartphone with navigation session already running.		
Assumptions:	The application is responsive and the stop control is visible to the user.		
Pre-Requisite:	User has already started a navigation session.		
Step No.	**Execution Description**	**Procedure Result**	
1	User taps the stop or exit navigation control.	Camera processing stops and the application exits navigation mode without crashing.	
Comments: Navigation session terminated correctly and cleanly.			
Passed			

7.3 Test Metrics

The following test metrics summarize the testing results of the implemented features of Sighteo. These metrics were used to evaluate the completeness and quality of the executed test cases. The majority of the tests were functional and were directly linked to the core use cases of the application, namely registration, login, and navigation assistance.

Table 7.9: Functional Test Case Metrics

Metric	Purpose / Value
Number of Test Cases	8
Number of Test Cases Passed	8
Number of Test Cases Failed	0
Test Case Defect Density	$(0 \times 100)/8 = 0$
Test Case Effectiveness	Since no defects were detected during the executed test cases, the value is recorded as 0 for this testing cycle.
Traceability Matrix	Each implemented feature was traced back to its corresponding use case and validated through one or more test cases, as shown in Table **??**.

In addition to the overall functional metrics, a small non-functional observation was also made regarding runtime behavior. During testing, the application remained responsive on supported Android devices and was able to start the navigation session, detect obstacles, and generate output without abrupt termination. This indicates that the implemented prototype is stable enough for controlled testing scenarios.

Table 7.10: Non-Functional Test Case Metrics

Metric	Purpose / Value
Application Stability	Stable during all executed test sessions
Navigation Session Crashes	0
Permission Handling Errors	0
Observed Runtime Behavior	Camera feed, obstacle detection, and feedback modules remained active during test execution
Purpose	To provide a concise non-functional view of prototype reliability during controlled testing

The Requirements Traceability Matrix presented herein describes the correspondence between system requirements, use cases and the corresponding test cases formulated during the testing phase. It ensures that each functional requirement of Sighteo system is duly implemented and verified through at least 1 test case. This mapping ensures that all major system features were investigated and no requirement left to be verified.

Table 7.11: Requirements Traceability Matrix

Req./Feature ID	Requirement / Feature	Use Case	Mapped Test Case(s)	Status
FR-01	User registration	Register User	TC-01	Verified
FR-02	User login	Login User	TC-02	Verified
FR-03	Camera initialization and permission handling	Navigate User	TC-03	Verified
FR-04	Real-time obstacle detection	Navigate User	TC-04	Verified
FR-05	Audio guidance output	Navigate User	TC-05	Verified
FR-06	Haptic feedback output	Navigate User	TC-06	Verified
FR-07	Safe direction suggestion	Navigate User	TC-07	Verified
FR-08	Navigation session stop/exit	Navigate User	TC-08	Verified

7.3.1 Conclusion

This chapter elaborates on the testing methodology used in order to validate the implemented components in the Sighteo system. A set of functional test cases were developed to evaluate key features such as user authentication, camera initialization, real-time obstacle detection, and combining audio and haptic guidance for navigation assistance. Each of the test cases was performed under controlled indoor conditions to make sure that the system would react to expected user interactions and environmental inputs. The results of the tests carried out show that the implemented modules are functioning as per the intended requirements and meet the functional requirements specified in the Software Requirements Specification. Test metrics and the Requirements Traceability Matrix further support the fact that all the major features of the system were tested through systematic testing. As a result of the testing process, it is possible to affirm that the developed prototype works reliably within the scope defined for the project.

Chapter 8 User Manual

The chapter is a detailed user guide to the Sighteo mobile application. The manual should serve as a guide to the user on the installation, access and usage of the main features of the system and as such, help the user operate it efficiently and exploit the navigation assistance capabilities effectively.

8.1 Getting Started

In order to start using the Sighteo application on the Android device, follow the steps as follows:

1. Install and download the Sighteo mobile application on the Android smartphone.
2. During installation, the application icon will appear on the device, and by tapping it, the application will be launched.
3. Upon opening the application, a welcome screen will be opened on which the user can log in or create a new account.
4. Ensure that the camera, speaker, and vibration permission of the device are turned on because they are mandatory in ensuring proper functionality of the application.

8.2 User Registration

In order to create a new account in the Sighteo application, follow the following procedure:

1. When you are on the welcome screen, locate the *Sign Up* button and tap it to get to the registration page.
2. To complete the form, fill in the necessary details namely, name, email address, and password in the respective fields.
3. Tap the *Register* button after filling the necessary details.
4. When the registration is successful, the system will enable the creation of an account and redirect the user to the login page.

8.3 User Login

In order to log into the application, do the following steps:

1. Start the Sighteo application.
2. On the log-in page, enter the registered email address and the password.

3. Click on the *Log In* button.
4. After signing in, the user will be redirected to the main dashboard.

8.4 Granting Camera Permission

Since Sighteo depends on real-time visual input to detect the obstacles, it is required that the camera permission be acquired. In the case where the first time is when the navigation feature is being launched, the application requests access to the camera.

1. Tap *Allow* to grant permission.
2. When the application is given permission, the camera is turned on and the live video feed is presented, which is required to detect objects.

8.5 Starting Navigation

In order to activate the navigation assistance feature, follow the following steps:

1. Upon logging in, the main screen of the application will appear.
2. Tap the *Start Navigation* button.
3. The application loads the camera feed and loads the object detecting model.
4. Put the device at the front of you so that the camera is able to detect the environment around.
5. The system initiates the process of detecting obstacles and real-time analysis of the environment.

8.6 Receiving Navigation Assistance

As one navigates, the application provides a user with audio and vibration feedback.

1. When the obstacle is detected, the system produces an audio message that notifies the user about the detected object.
2. The device can also send vibration messages when the user approaches an obstacle too closely.
3. The system could prescribe directional changes or move to the left or right side to avoid obstacles.
4. Care should be taken by the users, following the instructions given.

8.7 Stopping Navigation

In order to end the navigation, follow the following steps:

1. The mode of navigation is still on; however, find the *Stop Navigation* button on the screen.
2. Press the button to end the session.
3. The application stops processing of the camera and transfers the user back to the main screen.

8.8 Troubleshooting

In case the application does not work as expected, the following steps should be taken into account:

1. Make sure that you have camera permission.
2. Make sure that the device volume is on, which means that audio feedback can be heard.
3. MEnsure that the vibration facility is on.
4. Restart the application in case the navigation module fails to start.
5. Determine that the battery level of the device is sufficient to be used at a prolonged application.

8.9 Contact and Support

In case users face some problems during the use of the Sighteo application or need help, they can address the project team:

1. Click on the *Help* button located on the upper right corner of the screen.
2. Share specifics of the problem that was experienced.
3. The support team will analyze the request and provide support in time.

8.10 Conclusion

The chapter has outlined a step-by-step user manual to the Sighteo application explaining the steps required to install the application, create an account, log in, and use the navigation assistance capabilities. The guidelines below help to achieve proper functioning of the application and create an idea of how the system helps visually impaired people to find their way in the indoor settings safely.

Chapter 9 Experimental Results and Discussion

In this chapter we present the experimental evaluation of the three candidate object detection models used for our navigation system: YOLOv8, SSD and RetinaNet. The goal of the experiments is to understand the trade-off between detection accuracy and computational efficiency so that we can select a model that is reliable and practical for real-time use on a mobile device for visually impaired users.

9.1 Evaluation Setup

All three models were compared on a subset of the COCO validation set. The subset was carefully chosen to make sure it is sufficient to maintain a reasonable balance between reliability and computational cost for model comparison. The images are representative of indoor scenes with diverse objects common to visually impaired users, such as people, and daily items.

These experiments were run on a CPU, which more closely approximates the resource-constrained deployment environment of a smartphone than a high-end GPU. We measured a set of accuracy and efficiency metrics for each model.

The quality metrics of detection include the mean Average Precision at IoU 0.5, precision, recall, and F1-score. We have also recorded the error counts in terms of the number of true positives, false positives, and false negatives.

The efficiency metrics include the total processing time for the 200 images, the average inference time per image, frames per second (FPS), memory usage, number of parameters (in millions) and computational complexity in GFLOPs. In addition, we compute a derived efficiency score that combines accuracy and speed and is proportional to:

$$\text{Efficiency} = \text{FPS} \times \text{mAP0.5}_{\text{Number of Parameters.}}$$

9.2 Quantitative Comparison of Detection Accuracy

RetinaNet achieved the highest overall detection accuracy, with mAP@0.5 $\approx$ 0.49 and F1-score $\approx$ 0.56. It also attained the highest recall (about 0.66), meaning it missed the fewest objects and therefore produced the lowest number of false negatives. Its precision is moderate (about 0.48), which indicates more false positives than SSD.

SSD obtained moderate accuracy, with mAP@0.5 $\approx$ 0.31 and F1-score $\approx$ 0.50. Its precision is the highest of the three models (about 0.75), so when SSD predicts an object it is usually correct, but its recall is lower (about 0.37), meaning it still misses a noticeable fraction of objects.

YOLOv8 on this COCO subset produced a lower mAP@0.5 (around 0.06) and F1-score (around 0.33) compared to SSD and RetinaNet. Its recall is the lowest (around 0.28), so it misses more objects, and its precision is moderate (around 0.41). This is consistent with the fact that we are evaluating an off-the-shelf lightweight YOLOv8n model on a small validation subset without any fine-tuning.

From a purely accuracy-based perspective, RetinaNet clearly dominates, followed by SSD and then YOLOv8. However, for a visually impaired navigation aid, missing obstacles (false negatives) is more critical than raising extra alerts (false positives). On this safety-critical dimension, RetinaNet's high recall is particularly attractive.

9.3 Computational Efficiency Analysis

This corresponds to, respectively, about 4.6 FPS for YOLOv8, about 0.7 FPS for SSD and about 0.1 FPS for RetinaNet. YOLOv8 is thus an order of magnitude faster than SSD and more than 40 times faster than RetinaNet on CPU.

The total processing time for 200 images follows the same trend: YOLOv8 takes roughly 0.2 seconds to process the batch, SSD takes about 1.5 seconds and RetinaNet takes approximately 9.1 seconds. The time-breakdown plots show that most of this difference is due to the inference stage; preprocessing and post-processing are comparatively negligible.

Model size and computational complexity also differ significantly. YOLOv8 contains about 3.2 million parameters and requires around 8.7 GFLOPs. SSD contains roughly 35.6 million parameters with 62 GFLOPs, while RetinaNet has about 34.0 million parameters and almost 198 GFLOPs. YOLOv8 is therefore around ten times smaller than SSD and RetinaNet and requires far fewer floating-point operations, which directly translates into lower compute and energy usage on mobile hardware.

This gives an efficiency score of FPS $\times$ mAP@0.5/number of parameters of approximately $90,858$ for YOLOv8, about $6,020$ for SSD and roughly $1,586$ for RetinaNet. Although YOLOv8 has much lower mAP than many of the other methods, it achieves by far the highest overall efficiency score because it combines acceptable accuracy with extremely fast inference and a compact model size. In contrast, although RetinaNet is highly accurate, it is heavily penalized in this metric because its FPS is very low and its computational cost is correspondingly high.

These differences in efficiency are essential for a real-time navigation system that needs to run continuously on a smartphone. A model that cannot sustain near real-time FPS will feel laggy and may fail to warn users in time.

9.4 Class-wise Performance on COCO

To comprehend how each model behaves on different object types, we computed the average precision for the top ten COCO classes most relevant to navigation: person, bicycle, car, motorcycle, airplane, bus, train, truck, boat, and traffic light.

Among all classes, RetinaNet consistently has the highest average precision, often above 0.70 on some critical categories like person, motorcycle, airplane, and train. This confirms that the RetinaNet model is really strong in the detection of diverse object types when there is no constraint in terms of computation.

SSD is competitive on several classes, particularly where its high precision is helpful-for example, among vehicle categories, where there are a lot of false positives-but generally falls somewhat behind RetinaNet in average precision.

YOLOv8, while weaker on the aggregated mAP across all 200 images, still shows reasonably strong average precision for certain safety-critical classes, especially person, motorcycle, and train. That is, even with a lightweight configuration, YOLOv8 can still detect the most important obstacles for indoor and outdoor navigation with reasonable reliability, especially once fine-tuned on our custom dataset.

Overall, the class-wise analysis suggests that RetinaNet would be the best choice if maximum detection quality is required and hardware is powerful, while YOLOv8 provides a good baseline which can be further optimized for the classes most relevant for our use case.

9.5 Discussion and Model Selection for the Navigation System

The experiments demonstrate that each model performs a different balance of accuracy and efficiency, and a clear same trade-off between higher detection quality and practical, real-time performance on limited hardware. Details are provided as follows.

9.5.1 RetinaNet

RetinaNet provides the highest mAP@0.5, recall and F1-score and demonstrates the strongest class-wise average precision for the majority of categories. It is very well-suited for offline analysis or server-side processing, where the sole objective is to maximize the quality of detection and hardware resources are plentiful. However, the very slow throughput on CPU, high FLOPs, and big model size make this method unsuitable for on-device, real-time navigation.

9.5.2 SSD

SSD offers a reasonable middle ground: it reaches good precision and F1-score with a moderate mAP, while being simpler than the RetinaNet architecture. However, SSD still runs rather slow on CPU and is heavy in terms of parameters and GFLOPs, which makes further mobile use difficult without hardware acceleration.

9.5.3 YOLOv8

YOLOv8 is the fastest model, with approximately 4.6 FPS, the smallest number of parameters and the lowest FLOPs. It also achieves the highest efficiency score among the three candidates. Its main limitation is a lower mAP and F1-score in the out-of-the-box COCO evaluation and a lower recall than SSD and RetinaNet. Despite this, the class-wise results show that YOLOv8 still performs reasonably well on critical categories such as person and vehicle classes, and it leaves ample room for improvement through fine-tuning on a domain-specific indoor dataset and through threshold adjustment that favours higher recall.

9.5.4 Final Model Choice

In the context of a smartphone-based assistive navigation app for visually impaired users, latency, energy usage, and model size are as important as raw accuracy. A slightly-less-accurate model that runs reliably and fast on a mobile CPU is more useful in practice than a very accurate model responding several seconds late.

Based on experimental evidence from this chapter, YOLOv8 is picked for our prototype navigation system as the main detection backbone. In the subsequent chapter, we finetune YOLOv8 on our indoor dataset and integrate it into the pipeline of the mobile application, focusing on enhancing recall for safety-critical obstacle classes while preserving the real-time performance seen in this evaluation.

9.6 Summary of Comparative Analysis

This section summarizes the behavior of YOLOv8, SSD, and RetinaNet in terms of accuracy, speed, resource usage, and class-level precision in structured tables, hence providing readers with to-the-point insights regarding the trade-offs emerging during testing. It is an attempt to make this comparison easy to navigate, especially for diverse metrics influencing real-world model suitability.

9.6.1 Comprehensive Model Comparison

This work presents a summary of the performance of the models discussed above: YOLOv8, SSD, and RetinaNet. All results are tabulated to make it clear and easy to refer to; most key metrics have been brought out for each model in terms of accuracy, speed, and efficiency. The table format allows readers to rapidly scan and compare the performance of the models in a succinct and structured manner.

Table 9.1: Model Comparison Across All Metrics

Metric	YOLOv8	SSD	RetinaNet
mAP@0.5	0.0626	0.3147	0.4902
Precision	0.4108	0.7459	0.4776
Recall	0.2822	0.3719	0.6644
F1-Score	0.3346	0.4963	0.5557
TP/FP/FN	412/591/1048	543/185/917	970/1061/490
Total Time (s)	0.2152	1.4667	9.0881
Inference Time (s)	0.2078	1.4648	9.0865
FPS	4.65	0.68	0.11
Memory (MB)	0.0	0.0	2.9
Parameters (M)	3.2	35.6	34.0
FLOPs (G)	8.7	62.0	198.0
Power Score	2.78	220.98	673.50
Overall Score	0.492	0.315	0.556
Efficiency Score	90858.3	6019.8	1585.6

9.6.2 Class-wise Average Precision

This section shows the AP values of the models discussed above for the top ten classes in COCO. All results are presented in tabular form for clarity and ease of reference. The table highlights the category-level performance of each model. Such a layout will enable readers to compare, at once, the strengths and weaknesses of the detectors across different classes.

Table 9.2: Average Precision for Top 10 COCO Classes

Class	YOLOv8 AP	SSD AP	RetinaNet AP
person	0.618	0.457	0.705
bicycle	0.323	0.273	0.576
car	0.489	0.367	0.622
motorcycle	0.692	0.603	0.778
airplane	0.667	0.667	0.806
bus	0.377	0.269	0.529
train	0.600	0.600	0.760
truck	0.100	0.133	0.327
boat	0.143	0.000	0.143
traffic light	0.294	0.133	0.450

9.7 Conclusion

This chapter provided a thorough comparison between models-YOLOv8, SSD, and RetinaNet-using a common evaluation pipeline. The various differences in accuracy, speed, and efficiency were evident, with each model having a different strength. YOLOv8 had superior real-time performance and overall balance, so it was selected as the preferred model for further experimentation. SSD had moderate results, while RetinaNet achieved the highest accuracy with a significant computational cost. This helps in effective model choice for indoor object detection.

Chapter 10 Conclusions

Our final year project was to develop an obstacle-detecting assistive mobile application for visually impaired persons with real-time object detection using a smartphone's camera. The project was divided into two phases, namely FYP-1 and FYP-2, where the first one focused on the research and planning, and the second focused on the implementation and evaluation. The scope defined was accomplished and covered all the planned objectives of research, analysis, model selection and application development. Also, our focus was restricted to indoor environments only because the extension of the system to the outdoors would require significantly larger datasets and greater computational resources.

Our group laid the foundation for the project by conducting research, data analysis, and designing methodology throughout the first stage, i.e. FYP-I, of the Final Year Project. The major deliverables were:

- **Comprehensive Literature Review:** We conducted an extensive review of 15 research articles. This made us understand the existing solutions, identify technological gaps, and subsequently define the scope for our project in a concrete way.
- **Competitor Analysis:** We conducted a comprehensive evaluation of 7 existing applications and systems to understand their features, limitations, and market positioning that would help us in shaping a unique and practical solution.
- **Finalization of Methodology:** After carrying out much needed research and comparative studies, we came up with the overall methodology for our application which was compliant with our project objectives.
- **Model Comparison and Evaluation:** A full comparison between three object detection models: YOLO, SSD, and RetinaNet was conducted. Various performance metrics were calculated, which enabled us to identify the best model, YOLOv8, that would suit our application's requirements.

The second phase involved the **development and implementation** of the solution - an assistive application targeted at visually impaired people. It enabled the detection of obstacles by means of a phone's camera to guide users without the use of any other additional hardware. The main goals that the second phase, FYP-2, focussed on are:

- **Application Development:** Work on the actual mobile application development by following an Agile-based approach, which gave rise to flexibility and iteration for betterment.
- **Agile Development Process:** Split the work into sprints, continuously collected feedback, and improved the application based on the progress and testing results.

- **Model Training and Fine-Tuning:** Trained and fine-tuned the YOLOv8 model with the help of datasets of images of stairs, walls, and floors for better detection accuracy.
- **Testing and Validation:** Performed in-depth testing in a staging environment to make sure the application is stable, reliable, and responsive.
- **User Acceptance Testing:** Engaged visually impaired users in testing the application analyzing its practical usability and effectiveness.
- **Feedback Analysis:** Gathered and assessed users' suggestions to identify areas that could be further improved and refined the final application.

FYP-1 built the conceptual and technical foundations of our work, while FYP-2 transformed those into a functional application. Put together, these two phases provided an intelligent, accessible, and reliable assistive technology for the visually impaired.